
**Hierarchical structuring of weapon allometry in the
European earwig**

Journal:	<i>Evolution</i>
Manuscript ID	EVO-26-0122.R2
Manuscript Type:	Original Article
Keywords:	weapon, Allometry, environment, diet, social density, Sexual dimorphism

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Abstract

Static allometries describe how trait size covaries with body size and are often treated as stable descriptors of trait architecture. We quantified forceps allometry in the European earwig (*Forficula auricularia*) using a split-brood experiment manipulating diet and social density. Females, brachylabic males, and macrolabic males were treated as three levels of a phenotypic-group factor within a single hierarchical Bayesian model that allowed scaling relationships to vary among groups, environments, maternal families, and rearing dishes. Most posterior median slopes were below one. We found no clear evidence that female and brachylabic male slopes differed across environments, and macrolabic male slopes were also similar under most conditions. A steeper relationship was estimated for macrolabic males under poor diet at medium density, but the credible interval included isometry and the estimate was based on 18 individuals. Environmental differences in slopes were otherwise small or uncertain. A quadratic model received somewhat greater predictive support and suggested curvature near the extremes of the body-size distribution, particularly in macrolabic males, but did not alter the broad pattern of similarity. Variation among maternal families and rearing dishes was small. Overall, the results suggest that forceps allometry is broadly similar across phenotypes and environments, with limited and uncertain divergence associated with the exaggerated male morph.

Introduction

Static allometry describes how trait size covaries with body size among individuals at the same developmental stage and provides a central framework for understanding the evolution of exaggerated morphology. Sexually selected weapons often exhibit positive allometry, such that larger individuals tend to express disproportionately larger traits (Kodric-Brown et al. 2006; Bonduriansky 2007a; Voje 2016). However, hyperallometry is not universal. When traits serve both sexual and viability functions, or when functional and mechanical constraints limit disproportionate investment, isometry or hypoallometry may evolve instead (Bonduriansky and Day 2003; Eberhard et al. 2018; Kelly 2026). Consequently, scaling relationships reflect how developmental processes, functional performance, and selection interact to generate patterns of trait variation.

Static allometry is typically described by a linear relationship between log-transformed trait size and body size. In this framework, the slope (scaling exponent) describes the strength of covariation between trait size and body size and therefore the degree of size-dependent exaggeration (Pélabon et al. 2014; Voje 2016). Differences in scaling exponents among groups can indicate that trait exaggeration changes with body size, whereas similar slopes are consistent with a shared relationship between trait size and body size despite differences in average trait expression. Because scaling exponents link developmental variation in body size to variation in trait expression, they provide a useful means of evaluating how exaggerated traits vary within and among populations (Bonduriansky 2007a; Shingleton and Frankino 2013).

Sexual dimorphism is often viewed as the evolutionary divergence of males and females from a shared developmental architecture. However, dimorphism can arise through multiple pathways.

Males and females may differ in average trait expression while retaining similar allometric relationships, or divergence may occur through changes in how trait size scales with body size. Recent evidence from the maritime earwig *Anisolabis maritima* demonstrates that females, like males, can exhibit positively allometric forceps, highlighting that weapon exaggeration is not necessarily restricted to males (Matsuzawa and Konuma 2025). Species expressing alternative male phenotypes provide an opportunity to examine these processes in greater detail because males themselves are partitioned into discrete reproductive forms. In such systems, one male morph may resemble females more closely than other males, raising the possibility that sexual dimorphism is concentrated within a subset of the male population rather than distributed uniformly across all males. Understanding whether alternative male phenotypes differ from females in scaling relationships therefore offers insight into how sexual dimorphism and trait exaggeration evolve from shared developmental programs. Under the developmental threshold model, the major morph arises through activation of a developmental program for exaggerated trait growth when body size exceeds a critical value, whereas the minor morph retains a more “default” morphology produced in the absence of that program (Tomkins et al. 2005; Emlen et al. 2007); this framework provides a mechanistic basis for the classical view of males as exaggerated females (Darwin 1871; Bonduriansky 2007b) and predicts that the minor morph should resemble females more closely than the major morph in scaling relationships as well as in absolute trait size.

Species expressing alternative reproductive phenotypes provide particularly valuable models for examining the structure of allometry. In these species, males adopt discrete morphs that differ in weapon size, behaviour, and reproductive tactics (Gross 1996; Shuster and Wade 2003; Oliveira

et al. 2008). Because morphs share much of their genome and developmental environment, differences in scaling relationships can reveal how exaggeration evolves within a single species. Although alternative male phenotypes have been documented in many taxa, relatively few studies have quantified morph-specific static allometries, and even fewer have examined how developmental environments influence scaling relationships among morphs (Kochensparger et al. 2024; Kelly 2026).

Environmental conditions can influence both body size and trait expression and may therefore alter static allometries (Emlen 1997; Frankino et al. 2005; Shingleton et al. 2007). Nutritional stress, for example, can reduce investment in sexually selected structures, whereas social conditions such as rearing density may modify developmental trajectories through competition or social interactions. Environmental effects on scaling exponents would indicate plasticity in the relationship between trait size and body size, whereas similar scaling exponents across environments would imply canalization of allometric relationships (Frankino et al. 2005; Shingleton et al. 2007; Nijhout and German 2012). Determining whether developmental environments modify scaling relationships is therefore important for understanding the extent to which trait exaggeration is environmentally sensitive. Complementary structural modelling of the same study system demonstrates that developmental environments influence forceps expression both indirectly through body size and directly through trait-specific developmental pathways, particularly in males (Ozkan and Kelly submitted).

Beyond sex, morph, and environment, variation among maternal families can provide insight into the developmental and evolutionary structure of allometry. Families may differ not only in average trait expression but also in the strength of the relationship between trait size and body

size. Such variation could arise from genetic differences, maternal effects, or shared developmental conditions. Quantifying among-family variation in allometric slopes therefore offers a means of assessing whether scaling relationships themselves vary among related individuals (Lande 1980; Hansen and Houle 2008; Klingenberg 2008; Pélabon et al. 2014).

The European earwig (*Forficula auricularia*) provides a useful system for examining how sexual dimorphism and alternative male phenotypes are structured through allometry. Males express two discrete forceps morphs, commonly referred to as brachylabic and macrolabic males, which differ markedly in weapon size and shape (Simmons and Tomkins 1996; Tomkins et al. 2005) (Figure 1). Forceps function both as sexual weapons during male–male interactions and as defensive structures, making them traits with dual functions (Kamimura 2014; Meunier 2024).

Because these morphs coexist within the same population, earwigs provide an opportunity to ask whether males share a common scaling relationship that differs from females, or whether any divergence is associated more strongly with one male morph. Previous studies have documented morph-specific differences in forceps exaggeration, raising the possibility that alternative male phenotypes may follow different allometric trajectories (Simmons and Tomkins 1996; Tomkins et al. 2005). Moreover, recent work on another earwig species revealed positive allometry in female forceps, suggesting that sex differences in forceps scaling may be evolutionarily labile (Matsuzawa and Konuma 2025). In *F. auricularia*, the macrolabic morph bears the more exaggerated weapon, whereas the brachylabic morph is more common in many populations and tends to be favoured under poorer developmental conditions. This pattern is broadly consistent with the possibility that brachylabic males retain a more female-like morphology under a

threshold model of morph expression (Tomkins 1999; Tomkins and Brown 2004; Tomkins et al. 2005).

Here, we quantify static allometries of forceps length in female, brachylabid male, and macrolabid male *F. auricularia* reared under experimentally manipulated diet and density conditions. We first test whether the three phenotypic groups differ in allometric scaling relationships and whether any divergence is consistent across developmental environments. We then assess whether developmental conditions modify scaling exponents within each group and whether allometric relationships vary among maternal families. By integrating sex, morph, environment, and family effects within a hierarchical framework, we evaluate whether weapon allometry is broadly conserved across phenotypic groups or whether divergence is concentrated within particular male morphs.

Methods

Study species and experimental design

Adult female earwigs morphologically assignable to the *Forficula auricularia* species complex (*F. auricularia sensu lato*) were collected by hand at night from vegetation in Parc Outremont (Montréal, Québec, Canada) during September and October 2024. The European earwig is now recognized as a complex of closely related lineages that cannot always be distinguished reliably using external morphology alone (González-Miguéns et al. 2020). Because individuals were not genetically assigned to lineage, we refer to the study population as *F. auricularia sensu lato* here and use *F. auricularia* for readability elsewhere in the manuscript. This taxonomic uncertainty should not affect treatment comparisons in the present study because all individuals originated

from the same local population and were randomly assigned to experimental treatments, although caution may be warranted when comparing results among studies conducted on genetically characterized lineages.

Females ($n = 115$) were transported to the laboratory at the Université du Québec à Montréal and housed individually in 90-mm Petri dishes containing moist sand. Each dish was supplied with a cotton-plugged water tube and TetraMin Tropical Flakes (United Pet Group, Germany), and females were assigned unique identification codes. Earwigs were maintained in darkness at 10°C (Percival Scientific, Perry, IA, USA) to simulate overwintering conditions. Dishes were checked daily for egg laying, and food and water were replenished as required. Following oviposition (approximately early December), food and water were removed and dishes were inspected weekly. Beginning in January 2025, dishes were checked daily for hatching. Hatching occurred between 13 and 29 January 2025, and all 115 females successfully produced offspring.

For two weeks following hatching, families were provided with food and water ad libitum. This period allowed nymphs to complete an early moult and attain a size suitable for manual transfer to experimental dishes. To manipulate developmental conditions, we varied both nutritional quality and social density. Diet treatment consisted of either a good-quality diet (100% TetraMin Tropical Flakes) or a poor-quality diet (75% TetraMin Tropical Flakes and 25% non-digestible α -cellulose). Density treatment consisted of one, four, or eight individuals per dish. Offspring from each family were randomly assigned among treatment combinations using a split-brood design to minimize confounding between maternal background and environmental treatment. Experimental dishes were maintained at 20°C and contained a water tube, treatment-specific food, and a refuge constructed from four sections of paper straw. In total, 1,265 dishes were established. Mothers

were euthanized by freezing at -20°C after offspring had been assigned to treatments. Dishes were inspected several times per week for mortality and adult emergence, and food and water were replenished as needed.

Morphological measurements

Once offspring reached adulthood, they were removed from their dish, assigned a unique identification code, and the date of adult emergence was recorded. Pronotum length (a proxy for body size) and forceps length were photographed under a Leica S6D stereomicroscope (Leica Microsystems Inc., Concord, ON, Canada) at $0.68\times$ magnification using Enersight software. Measurements were obtained to the nearest 0.001 mm using Fiji image-analysis software (Schindelin et al. 2012) following established procedures (van Lieshout and Elgar 2009). Measurements were conducted blind to treatment whenever possible. To assess measurement repeatability, a subset of 30 individuals was measured twice. Intraclass correlation coefficients indicated high repeatability for both pronotum length ($\text{ICC} = 0.94$) and forceps length ($\text{ICC} = 0.97$).

Data preparation and statistical analyses

All analyses were conducted in R version 4.4.2 (R Core Team 2025). Individuals were retained only when the final number of survivors preserved the intended density treatment (low density: one survivor; medium density: at least three survivors; high density: at least six survivors). Individuals lacking sex information were excluded. Final sample sizes for each phenotypic group and developmental environment are provided in Table S1. Pronotum length and forceps length were log-transformed prior to analysis to linearize allometric relationships and reduce heteroscedasticity.

Male morphs were classified using finite mixture modelling implemented in the package *mixsmsn* (Prates et al. 2013). Raw male forceps lengths were modelled as a two-component skew-normal mixture distribution, and males were assigned to the component with the highest posterior probability. This procedure identified the two discrete male morphs traditionally recognized in *F. auricularia*: brachylabic and macrolabic males.

We analysed allometric relationships using a hierarchical Bayesian model fitted with the package *brms* (Bürkner 2017) and the *cmdstanr* backend. The final analysis included 1,914 individuals from 110 maternal families and 667 rearing dishes. Log-transformed forceps length was modelled as a function of log-transformed pronotum length (body size), phenotypic group, diet, density, and all lower-order and higher-order interactions among these predictors. Phenotypic group comprised three levels: females, brachylabic males, and macrolabic males. Treating females and the two male morphs as equivalent levels of a single factor allowed direct comparisons among all phenotypes without assuming a single male allometric relationship.

Using descriptive variable names, the primary linear model can be represented in R notation as:

$$\text{logF} \sim \text{logP} * \text{group} * \text{diet} * \text{density} + (1 + \text{logP} \mid \text{mother_id}) + (1 \mid \text{rearing_dish})$$

where logF is log-transformed forceps length, logP is log-transformed pronotum length, group is phenotypic group, mother_id identifies maternal family, and rearing_dish identifies the rearing dish. Interactions involving body size allowed allometric slopes to vary among phenotypic groups and developmental environments, whereas terms not involving body size described differences in model intercepts among groups and environments. Maternal family was

included as a varying intercept and varying slope for body size, allowing families to differ in both average forceps expression and allometric scaling. Rearing-dish identity was included as a varying intercept to account for non-independence among individuals raised in the same dish.

To evaluate whether allometric relationships deviated from linearity, we fitted an alternative model that included a quadratic body-size term and its interactions with phenotypic group, diet, and density. This model was specified as:

$$\log F \sim (\log P + I(\log P^2)) * \text{group} * \text{diet} * \text{density} + (1 + \log P \mid \text{mother_id}) + (1 \mid \text{rearing_dish})$$

Predictive performance of the linear and quadratic models was compared using approximate leave-one-out cross-validation (LOO) (Vehtari et al. 2017). Because the primary objective was to estimate interpretable condition-specific allometric exponents, the linear model was used to summarize average slopes, whereas the quadratic model was treated as a sensitivity analysis for localized departures from linearity.

Weakly informative priors were specified as normal(0, 1) for fixed-effect coefficients, normal(0, 2) for the intercept, Student- $t(3, 0, 1)$ for residual and varying-effect standard deviations, and LKJ(2) for the maternal intercept–slope correlation. Models were fitted using four Hamiltonian Monte Carlo chains of 4,000 iterations each, including 2,000 warm-up iterations, with an adapt_delta of 0.97. Convergence was assessed using trace plots, effective sample sizes, divergent transitions, and the potential scale reduction statistic [$\hat{R}$; Vehtari et al. (2021)]. Both models showed satisfactory convergence (Table S2).

Condition-specific allometric slopes were estimated from the posterior distribution of the linear model using the function `emtrends()` in the package *emmeans* (Lenth et al. 2019). Slopes were estimated separately for each combination of phenotypic group, diet treatment, and density treatment. Pairwise contrasts among phenotypic groups, diet treatments, and density treatments were derived from the same posterior distributions. Contrasts were calculated as the first level minus the second level—for example, female minus macrolabid male, good minus poor diet, or low minus medium density—such that negative values indicate a larger slope estimate for the second level. We report posterior medians and 95% credible intervals throughout. Inference was based on posterior distributions rather than null-hypothesis significance testing.

Results

Females exhibited a continuous distribution of weapon sizes across body sizes (pronotum length: mean $\pm$ SD = 1.77 ± 0.14 mm; forceps length: mean $\pm$ SD = 3.59 ± 0.24 mm, $n = 935$). Males formed two discrete clusters corresponding to brachylabid (pronotum length: mean $\pm$ SD = 1.66 ± 0.13 mm; forceps length: mean $\pm$ SD = 4.06 ± 0.30 mm, $n = 491$) and macrolabid morphs (pronotum length: mean $\pm$ SD = 1.80 ± 0.12 mm; forceps length: mean $\pm$ SD = 5.93 ± 0.72 mm, $n = 488$), with overlap in body size but substantial separation in forceps length (Figure 2).

Differences among females and male morphs

Forceps length generally increased with pronotum length, and most posterior median slopes were below one (Figure 3; Table S5; Figure S1). Treating females, brachylabid males, and macrolabid males as three phenotypic groups revealed broadly similar scaling relationships across most developmental environments. Under good diet and low density, estimated slopes were 0.37 (95%

236 CrI: 0.18–0.55) in females, 0.43 (95% CrI: 0.20–0.66) in brachylabial males, and 0.46 (95% CrI:
237 0.21–0.71) in macrolabial males. Pairwise differences among groups were generally small or
238 uncertain, with contrast intervals overlapping zero across most environments (Table S6).

239 A quadratic model had somewhat better expected out-of-sample predictive performance than the
240 linear model ($\Delta\text{ELPD} = 13.29$, $\text{SE} = 6.68$; Table S4), providing some evidence of curvature in the
241 relationship between body size and forceps length. The quadratic predictions did not substantially
242 alter the broad comparison among phenotypic groups, although they revealed localized
243 departures from linearity (Figure S2). Curvature was most apparent near the extremes of the
244 observed body-size range in several macrolabial male groups, where predictions were influenced
245 by relatively few individuals. These edge patterns should therefore be interpreted cautiously and
246 do not necessarily imply strong curvature across the central portion of the body-size distribution.
247 We use the linear model to summarize average condition-specific allometric exponents and treat
248 the quadratic model as a sensitivity analysis.

249 Across the full set of developmental environments, there was little evidence that females and
250 brachylabial males differed consistently in allometric slope (Table S6). The contrast interval
251 between females and brachylabial males included zero in every diet-by-density combination
252 examined. Macrolabial male slopes were also similar to those of females and brachylabial males
253 under most environmental conditions. The clearest exception occurred under poor diet at medium
254 density, where the estimated slope for macrolabial males (1.23, 95% CrI: 0.77–1.73) was steeper
255 than those estimated for females (0.33, 95% CrI: 0.15–0.49) and brachylabial males (0.26, 95%
256 CrI: 0.04–0.49). Pairwise contrast intervals excluded zero in this environment (Table S6). This

result should nevertheless be interpreted cautiously because the estimate was based on only 18 poor-diet macrolabid males at medium density (Table S1).

Posterior median slopes for females and brachylabid males were below one in every developmental environment examined. The posterior median for macrolabid males exceeded one only under poor diet at medium density (1.23, 95% CrI: 0.77–1.73). Because this credible interval included one, the evidence that this relationship was hyperallometric rather than isometric was uncertain. The small sample of poor-diet macrolabid males further limits inference from this treatment combination (Table S1).

Environmental effects on scaling

Estimated environmental differences in allometric slopes were generally small or uncertain. In females, diet contrast intervals included zero at all densities (Figure 3; Table S7), providing little evidence that nutritional conditions altered scaling relationships. Diet contrasts for brachylabid males were likewise uncertain across all density treatments (Table S7).

The clearest indication of environmental variation involved macrolabid males, but these estimates should be interpreted cautiously because poor-diet macrolabid males were uncommon at low density ($n = 3$), medium density ($n = 18$), and high density ($n = 13$) (Table S1). At medium density, the estimated slope was steeper under poor diet than under good diet (good–poor contrast = -0.81, 95% CrI: -1.33 to -0.29; Table S7). Diet contrast intervals included zero at low and high densities (Table S7).

Density contrast intervals also included zero for females and brachylabid males (Figure 3; Table S8). Among poor-diet macrolabid males, the estimated slope at medium density exceeded those at

low density (low–medium contrast = -1.26, 95% CrI: -2.26 to -0.29) and high density (medium–high contrast = 1.20, 95% CrI: 0.29 to 2.02; Table S8). Thus, the clearest estimated environmental differences were associated with poor-diet macrolabial males at medium density. Because all three poor-diet macrolabial groups had small samples, these contrasts should be regarded as provisional.

Among-family variation

Estimated variation among maternal families was small (Table S3). The posterior median standard deviation of family-level intercepts was 0.02 (95% CrI: 0.00–0.07), indicating limited among-family variation in forceps expression after accounting for body size, phenotypic group, diet, and density. The standard deviation of family-level allometric slopes was 0.05 (95% CrI: 0.02–0.14), indicating some uncertainty and variation among families in the relationship between body size and forceps length.

The correlation between family-level intercepts and slopes was negative but highly uncertain ($r = -0.79$, 95% CrI: -0.99–0.58), providing little evidence for a consistent association between forceps expression and allometric scaling among families. Variation among rearing dishes was small (SD = 0.01, 95% CrI: 0.00–0.02), as was residual variation (SD = 0.07, 95% CrI: 0.07–0.08).

Discussion

Static allometries are often treated as stable descriptors of morphological architecture (Huxley 1932; Gould 1966; Voje 2016), yet the extent to which scaling relationships vary among phenotypes and environments remains an important question in evolutionary biology (Pélabon et

al. 2014; Voje 2016). By analysing females, brachylabial males, and macrolabial males as separate phenotypic groups, we found that forceps allometry in the European earwig was characterised more by similarity than by consistent divergence. There was no clear evidence that females and brachylabial males differed in slope in any developmental environment, and macrolabial male slopes were also similar to those of the other groups under most conditions. Where a clear slope difference was estimated, it involved macrolabial males under poor diet at medium density, a treatment combination represented by only 18 individuals. Environmental contrasts were otherwise generally small or uncertain. These results suggest that forceps allometry is broadly similar across phenotypes and environments, while leaving open the possibility of limited, context-dependent divergence associated with the exaggerated male morph.

These findings differ quantitatively from two influential earlier analyses of *F. auricularia*. In two field populations from the Farne Islands, Tomkins and Simmons reported least-squares slopes of 0.68–0.70 for brachylabial males and 1.70–1.94 for macrolabial males, together with still steeper reduced-major-axis estimates, and concluded that the morphs differed in both slope and elevation (Tomkins and Simmons 1996). Tomkins et al. subsequently reported reduced-major-axis slopes of 1.15 for minor males, 3.95 for all major males, and 1.78 for asymptotic major males, interpreting the broken log–log relationship as evidence of size-dependent reprogramming (Tomkins et al. 2005). Our data likewise recover discrete male forceps modes, extensive overlap in body size, and a pronounced difference in average forceps expression, but they do not consistently recover steep macrolabial slopes: posterior medians were below 1 in nearly all environments, and morph contrasts were usually uncertain. Across studies, therefore,

320 discontinuity in forceps expression appears more repeatable than strong divergence in within-
321 morph allometric slope.

322 Direct numerical comparisons should nevertheless be made cautiously. The earlier studies
323 analysed field-collected males from UK island populations, used pronotum width as the body-
324 size measure, and classified morphs by forceps shape or forceps-length cutoffs; they also relied
325 on least-squares or reduced major axis regressions. We analysed laboratory-reared offspring from
326 a Montréal population of *F. auricularia sensu lato*, used pronotum length, classified males using
327 mixture modelling of forceps length, and estimated conditional slopes within developmental
328 environments using a hierarchical Bayesian regression. In particular, reduced-major-axis slopes
329 are not directly equivalent to the conditional regression slopes estimated here. Even with these
330 differences, the qualitative contrast is substantial and suggests that steep macrolabial
331 hyperallometry may vary among populations, lineages, or developmental contexts, or may be
332 sensitive to how morphs and body size are defined.

333 Sexual dimorphism is classically framed as the divergence of males from a shared female
334 “ground plan,” with males viewed as modified females and exaggerated male traits interpreted as
335 departures from a default developmental programme (Darwin 1871; Bonduriansky 2007b). Under
336 this view, alternative male morphs might differ from females in proportion to their degree of
337 exaggeration. Our results provide only limited support for this expectation: brachylabial male
338 slopes did not clearly differ from female slopes, whereas the clearest estimated departure
339 involved macrolabial males in one sparsely sampled developmental environment. The principal
340 distinction between the male morphs in our data was therefore one of forceps elevation and
341 absolute expression rather than a consistently different scaling exponent. This distinction does

not rule out a developmental threshold or reprogramming event, because such a transition could alter the elevation or timing of weapon growth without generating a uniformly steeper adult static allometry in every environment. However, because morphs were classified from the adult forceps distribution, the present analysis cannot independently demonstrate a developmental switch.

Threshold models in horned beetles have likewise proposed reduced activation of weapon-growth pathways in minor males, although the extent to which alternative morphs are developmentally uncoupled remains debated (Tomkins et al. 2005; Emlen et al. 2007). Comparative evidence from neotropical harvestmen suggests that minor-male weapon expression may track female expression more closely than major-male expression across species (Buzatto et al. 2014).

Whether a comparable developmental mechanism underlies the pattern in *F. auricularia* will require direct study of morph determination and forceps growth.

Morph-specific scaling does not follow a single pattern across male-dimorphic systems. In the Wellington tree wētā (*Hemideina crassidens*), weapon allometries did not align closely with the degree of weapon exaggeration or the presumed mating role of each morph (Kelly 2026). The present estimates provide a tentative contrast because the exaggerated macrolabic morph exhibited the clearest allometric departure, though only under one developmental condition.

Taken together, these studies suggest that alternative reproductive phenotypes can differ substantially in weapon expression without necessarily showing consistent differences in static allometry. The relationship between morph exaggeration and allometric slope may therefore depend on the developmental and functional context of each system.

The prevalence of posterior median slopes below one is also noteworthy. Sexually selected weapons are often expected to exhibit positive or hyperallometric scaling (Kodric-Brown et al.

2006; Bonduriansky 2007a), yet the posterior median slope was below one in all but one phenotype-by-environment combination. In the remaining combination, the median exceeded one, but its credible interval included isometry. This pattern is consistent with growing recognition that exaggerated sexual traits need not be hyperallometric (Bonduriansky and Day 2003; Eberhard et al. 2018; Kelly 2026) and with comparative work showing that allometric slope and intercept can evolve partly independently (Painting et al. 2024). Earwig forceps serve multiple functions, including male–male competition, predator defence, prey handling, and courtship interactions (Kamimura 2014; Meunier 2024). Such multifunctionality could limit disproportionate investment in a single function and favour more moderate scaling, although the present study does not test that mechanism directly. A recent study in *F. auricularia* also found no association between male forceps length and six behavioural or physiological traits commonly assumed to indicate male quality (Blackwell et al. 2024), suggesting that the selective information conveyed by forceps size may be less straightforward than a classical condition-dependent weapon model assumes. The broadly similar slopes estimated for females and both male morphs are consistent with shared functional or developmental constraints despite marked differences in absolute forceps size. Positive allometry in female *Anisolabis maritima* (Matsuzawa and Konuma 2025) further illustrates that female forceps scaling varies among earwig species and should not be assumed to follow a single regime.

Environmental effects on scaling relationships were generally limited or imprecisely estimated. Diet and density contrast intervals included zero for females and brachylabic males, and the clearest environmental contrasts were confined to poor-diet macrolabic males at medium density. Because poor-diet macrolabic males were uncommon in all density treatments, these estimates

are sensitive to small samples and should be treated as provisional. The quadratic model had somewhat better expected predictive performance than the linear model and revealed localized curvature in several relationships, especially near the extremes of the body-size distribution in macrolabial males. Those portions of the fitted curves were influenced by relatively few observations, however, and did not provide strong evidence of curvature across the central body-size range. We therefore interpret the linear slopes as average summaries of condition-specific scaling rather than complete descriptions of every relationship. Overall, the data provide little evidence for broad environmental restructuring of allometry, while suggesting a possible environment-specific response in the exaggerated morph that warrants targeted testing with more balanced samples. This interpretation is also consistent with evidence that selection on weapon allometry can act more strongly on the intercept than on the slope (Embets et al. 2023), raising the possibility that developmental environments influence weapon expression primarily through average trait size rather than through large changes in scaling.

The similarity of female and brachylabial male slopes may provide some insight into morph differentiation, but it does not identify the underlying mechanism. Morph expression in *F. auricularia* is influenced by both genetic and environmental factors (Tomkins 1999), and morph frequencies can shift with population density over short geographic distances (Tomkins and Brown 2004). The substantial overlap in body size between brachylabial and macrolabial males in our sample mirrors the earlier field studies and confirms that body size alone does not perfectly separate the morphs (Tomkins and Simmons 1996; Tomkins et al. 2005). At the same time, the combination of discrete forceps modes and generally similar within-morph slopes suggests that the main divergence in our population lies in the elevation of forceps expression rather than in

wholesale restructuring of the relationship between body size and forceps length. This pattern is compatible with several developmental scenarios, including a threshold-like shift in weapon growth, but adult static allometries alone cannot distinguish reprogramming from continuous growth processes, changes in growth duration, or resource constraints (Tomkins et al. 2005; Emlen et al. 2007). Direct measurements across juvenile development would be needed to identify which mechanism produces the observed morph difference.

Estimated variation among maternal families was small, although the posterior distribution for the family-level slope standard deviation indicated some heterogeneity in scaling among families. The correlation between family-level intercepts and slopes was highly uncertain, providing little evidence that families producing relatively large forceps also tended to show steeper allometries. Because maternal identity combines genetic effects, maternal effects, and aspects of the shared early environment, these data cannot establish that the observed family variation is heritable. They instead indicate that scaling relationships may vary modestly among maternal lineages and motivate designs capable of separating genetic from environmental sources of that variation (Lande 1980; Pélabon et al. 2014).

More broadly, our results are compatible with viewing static allometry as a hierarchical phenotype shaped by variation at several biological levels (Emlen and Nijhout 2000). The most robust pattern in this study was broad similarity across females, male morphs, and developmental environments, accompanied by a limited number of context-specific departures. Analysing the male morphs separately was nevertheless important because pooling them would have concealed the steeper slope estimated for poor-diet macrolabid males at medium density and the greater curvature near the extremes of some macrolabid body-size distributions. Studies that pool

alternative male phenotypes may therefore overlook heterogeneity in scaling, even when that heterogeneity is subtle or restricted to particular environments (e.g. Kelly 2005; see also Kochensparger et al. 2024). Our findings support treating alternative reproductive phenotypes explicitly in analyses of sexual dimorphism, while also emphasizing that differences among morphs should not be assumed to be pervasive. When considered alongside the earlier UK studies, the present results further suggest that steep major-morph hyperallometry is not necessarily an invariant species-level property but may depend on population, lineage, developmental environment, morph definition, or analytical scale. In *F. auricularia*, forceps allometry appears broadly similar across phenotypes and environments, with tentative evidence that the exaggerated male morph may deviate from this shared pattern under certain developmental conditions.

Data availability

Data are available through the OSF repository at https://osf.io/m7c53/overview?view_only=ce50e42b29284cbe8164e768ecb9c2b1.

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551 **Figures**

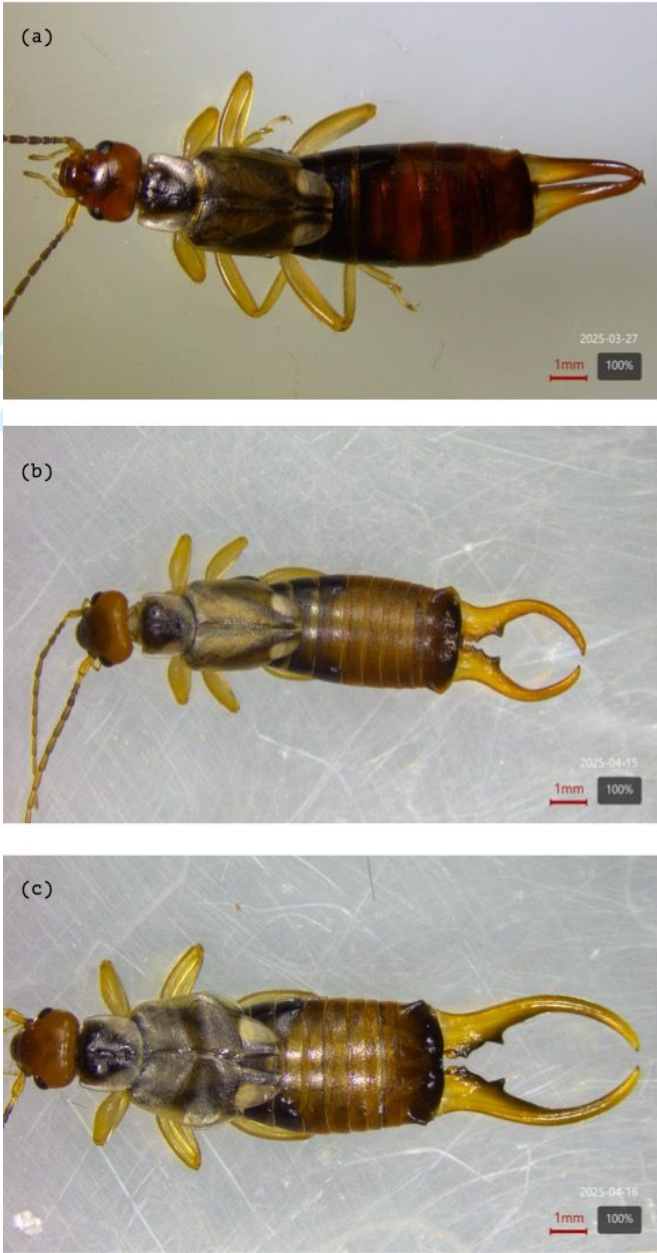


Figure 1: Visual comparison of forceps size and shape across sexes and male morphs: (a) female, (b) brachylabic male, and (c) macrolabic male. Macrolabic males have longer and more strongly curved forceps than brachylabic males and females. Photographs are of adults from the experimental population (Parc Outremont, Montréal). Scale bars = 1 mm.

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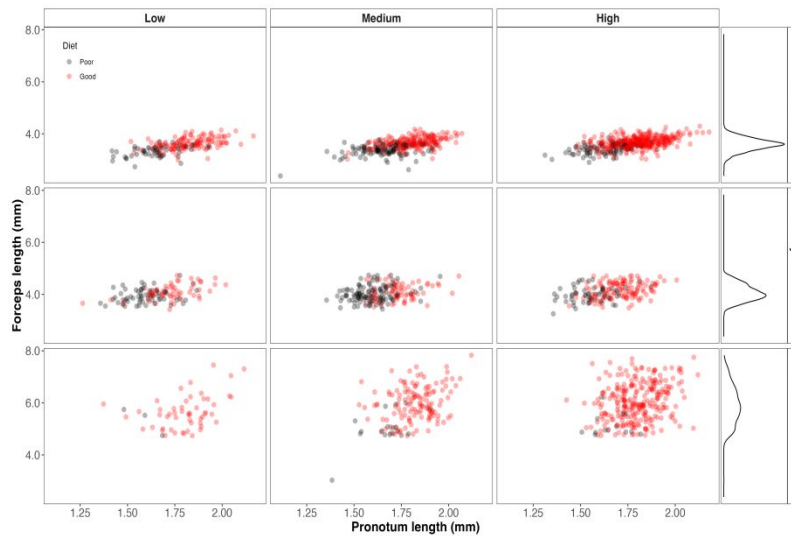


Figure 2: Relationship between pronotum length (body size) and forceps length in female, brachylabial male, and macrolabial male *Forficula auricularia* reared under two diet treatments (good and poor) and three density treatments (low, medium, and high). Points represent individual adults. Rows correspond to phenotypic groups and columns correspond to density treatments. Colours indicate developmental diet. Density distributions of forceps length for each phenotypic group are shown in the right-hand panels. Females exhibited a continuous distribution of forceps sizes, whereas males formed two discrete morphs that overlapped in body size but differed markedly in forceps length. Macrolabial males possessed substantially longer forceps than brachylabial males across the observed range of body sizes. Sample sizes for each diet-by-density combination are provided in Table S1.

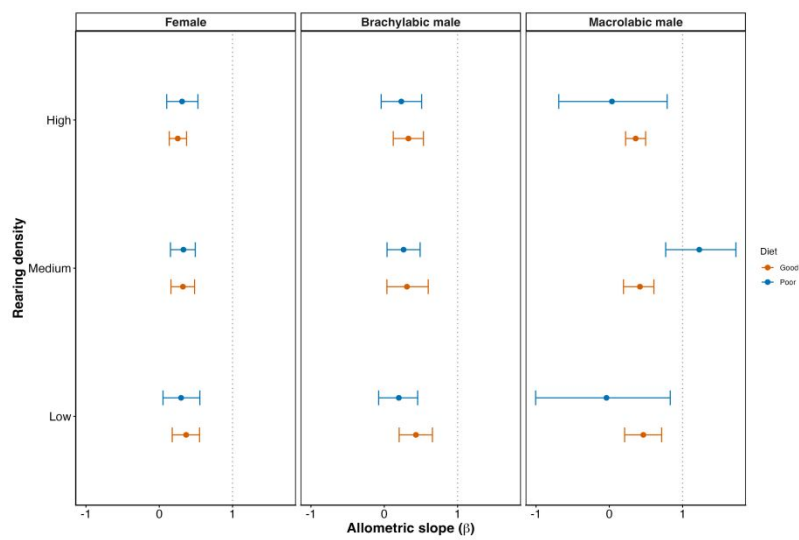
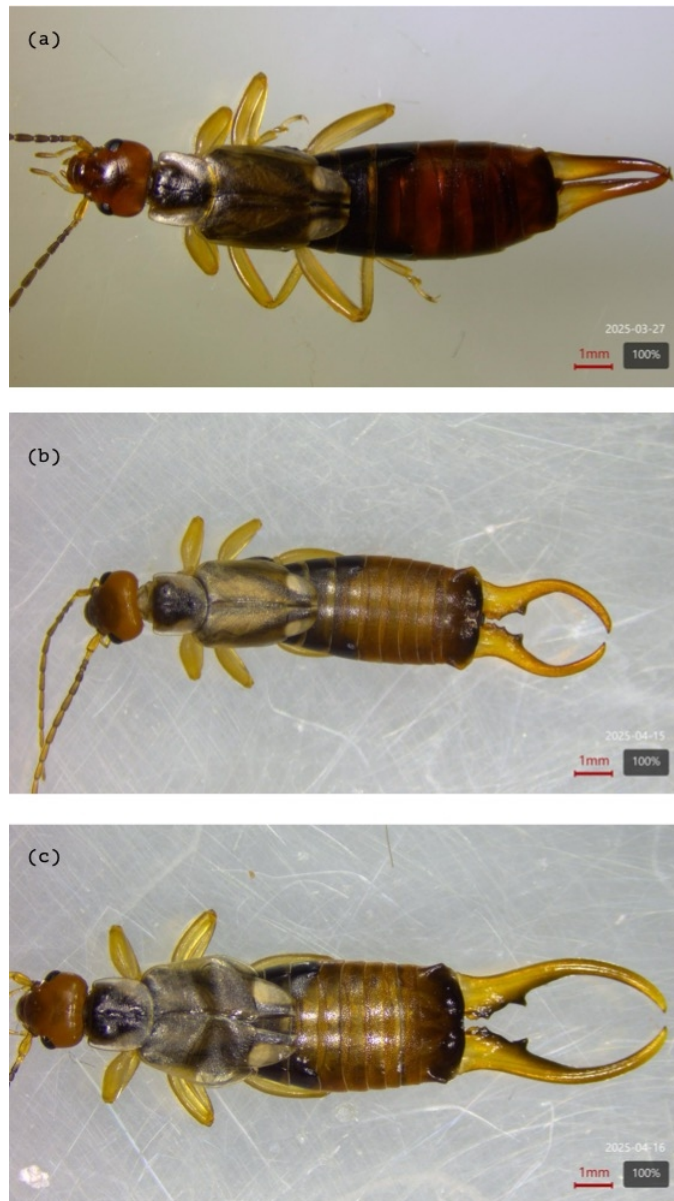
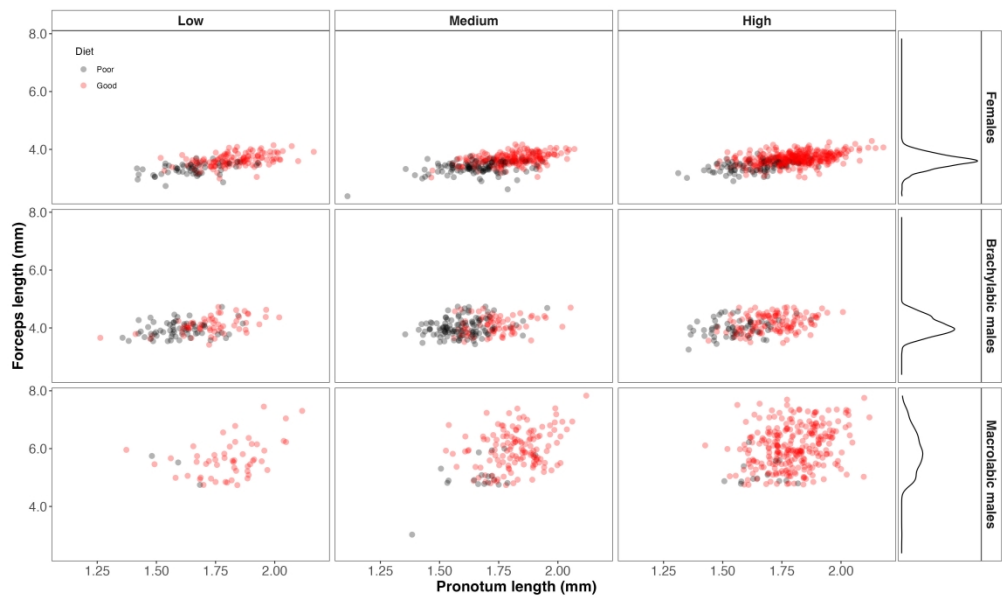


Figure 3: Posterior median allometric slopes (β) and 95% credible intervals describing the relationship between forceps length and pronotum length in females, brachylabic males, and macrolabic males across diet and density treatments. Slope estimates were derived from a single Bayesian mixed-effects model fitted to all individuals, with phenotypic group, diet, density, and their interactions with body size included as fixed effects. The vertical dashed line indicates isometry ($\beta = 1$). Posterior medians below or above one correspond to hypoallometric or hyperallometric point estimates, respectively; credible intervals indicate uncertainty relative to isometry.



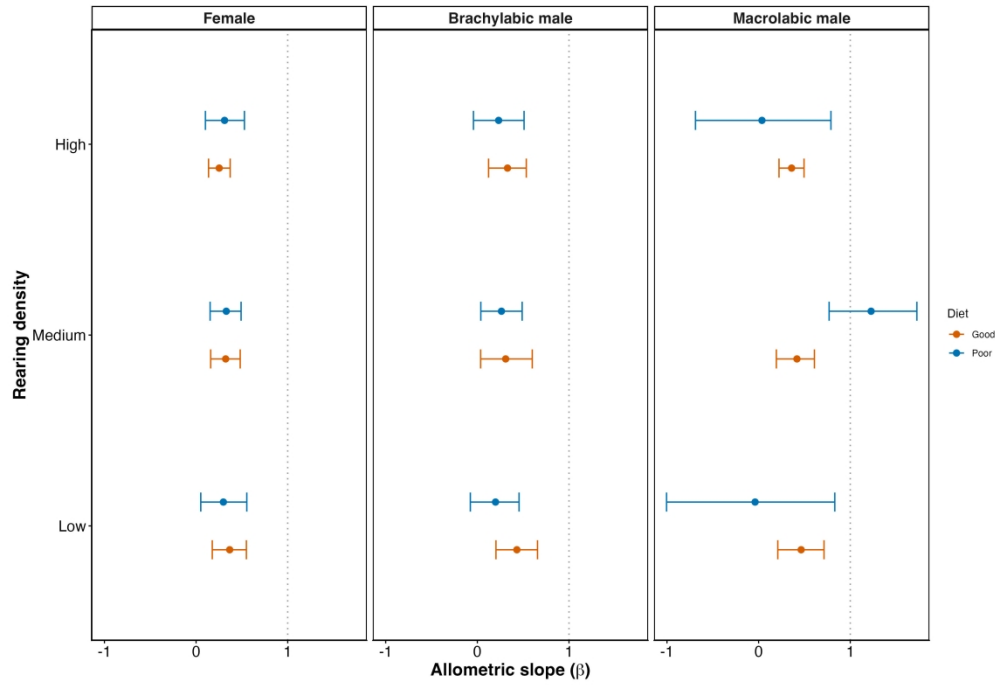
Visual comparison of forceps size and shape across sexes and male morphs: (a) female, (b) brachylabic male, and (c) macrolabic male. Macrolabic males have longer and more strongly curved forceps than brachylabic males and females. Photographs are of adults from the experimental population (Parc Outremont, Montréal). Scale bars = 1 mm.

199x352mm (72 x 72 DPI)



Relationship between pronotum length (body size) and forceps length in female, brachylabic male, and macrolabic male *Forficula auricularia* reared under two diet treatments (good and poor) and three density treatments (low, medium, and high). Points represent individual adults. Rows correspond to phenotypic groups and columns correspond to density treatments. Colours indicate developmental diet. Density distributions of forceps length for each phenotypic group are shown in the right-hand panels. Females exhibited a continuous distribution of forceps sizes, whereas males formed two discrete morphs that overlapped in body size but differed markedly in forceps length. Macrolabic males possessed substantially longer forceps than brachylabic males across the observed range of body sizes. Sample sizes for each diet-by-density combination are provided in Table S1.

1569x934mm (72 x 72 DPI)



Posterior median allometric slopes (β) and 95% credible intervals describing the relationship between forceps length and pronotum length in females, brachylabid males, and macrolabid males across diet and density treatments. Slope estimates were derived from a single Bayesian mixed-effects model fitted to all individuals, with phenotypic group, diet, density, and their interactions with body size included as fixed effects. The vertical dashed line indicates isometry ($\beta = 1$). Posterior medians below or above one correspond to hypoallometric or hyperallometric point estimates, respectively; credible intervals indicate uncertainty relative to isometry.

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Hierarchical structuring of weapon allometry in the European earwig.

For Peer Review Only

Abstract

Static allometries describe how trait size covaries with body size and are often treated as stable descriptors of morphological trait architecture, ~~yet scaling relationships themselves may evolve and respond plastically to environmental conditions.~~ We quantified ~~sex- and morph-specific weapon allometries (focusing on the forceps, which are hardened rear appendages used for defence, hunting, and mating)-~~ allometry in the European earwig (*Forficula auricularia*) ~~under experimental manipulation of developmental~~ using a split-brood experiment manipulating diet and social density ~~using hierarchical models that allow scaling exponents to vary across biological levels.~~ Forceps length scaled positively with body size in all groups, but males exhibited substantially steeper allometric slopes than females. There was some evidence that ~~nutritional stress was associated with reduced male scaling exponents, although support for these effects was variable and context-dependent, while density had little consistent influence on slopes.~~ Female scaling relationships remained comparatively stable across environments. Among males, ~~macrolabic individuals tended to show greater environmental sensitivity in scaling relationships than.~~ Females, brachylabic males, although statistical support for these differences was limited. There was little evidence that environmental or morph-specific differences ~~and~~ macrolabic males were driven by intercept shifts independent of body size. Among family ~~variation in~~ treated as three levels of a phenotypic-group factor within a single hierarchical Bayesian model that allowed scaling relationships to vary among groups, environments, maternal families, and rearing dishes. Most posterior median slopes were below one. We found no clear evidence that female and brachylabic male slopes differed across environments, and macrolabic male slopes and intercepts was modest, were also similar under most conditions. A steeper

26 relationship was estimated for macrolabic males under poor diet at medium density, but the
27 credible interval included isometry and their covariance was weak and the estimate was based on
28 18 individuals. Environmental differences in slopes were otherwise small or uncertain. A
29 quadratic model received somewhat greater predictive support and suggested curvature near the
30 extremes of the body-size distribution, particularly in macrolabic males, but did not alter the
31 broad pattern of similarity. Variation among maternal families and rearing dishes was small.
32 Overall, ~~thesethe~~ results suggest that ~~weaponforceps~~ allometry is ~~structured primarily through~~
33 ~~variation in scaling exponents and provide qualified support for the idea that slopes may~~
34 ~~represent condition-sensitive components of broadly similar across phenotypes and~~
35 ~~environments, with limited and uncertain divergence associated with the exaggerated trait~~
36 ~~architecture~~male morph.

37

Introduction

Static allometry describes how trait size covaries with body size among individuals at the same developmental stage and provides a central framework for understanding the evolution of exaggerated morphology. Sexually selected weapons often exhibit positive allometry, such that larger individuals tend to express disproportionately larger traits (Kodric-Brown et al. 2006; Bonduriansky 2007a; Voje 2016). However, hyperallometry is not universal. When traits serve ~~dual~~both sexual and viability functions, or when functional and mechanical constraints limit disproportionate investment, isometry or hypoallometry may evolve instead (Bonduriansky and Day 2003; Eberhard et al. 2018; Kelly 2026). Consequently, scaling relationships reflect how developmental processes, functional performance, and selection ~~are associated with~~interact to generate patterns of trait variation.

Static allometry is typically described by a linear relationship between log-transformed trait size and body size. In this framework, the slope (scaling exponent) describes the strength of covariation between trait size and body size (~~i.e., and therefore the degree of~~ size-dependent exaggeration), ~~whereas the intercept (elevation) represents expected trait size at a reference value of body size (e.g., the mean when body size is centred). Static allometry can therefore diverge through changes in either parameter. (Pélabon et al. 2014; Voje 2016).~~ Differences in scaling exponent ~~reflect variation in the strength of covariation~~exponents among groups can indicate that trait exaggeration changes with body size, whereas similar slopes are consistent with a shared relationship between trait size and body size, ~~whereas intercept divergence reflects despite~~ differences in ~~expected trait size at that reference body size. These components need not evolve together (Lande 1980; Cheverud 1982; Pélabon et al. 2014).~~ Sexual dimorphism, for example,

may arise because males and females differ in scaling sensitivity, in expected average trait expression. Because scaling exponents link developmental variation in body size to variation in trait size, or in both. Similarly, alternative male morphs may diverge through elevation shifts while retaining similar scaling exponents, or through differences in scaling relationships (Tomkins et al. 2005). Distinguishing between slope-based and intercept-based divergence is therefore essential for understanding expression, they provide a useful means of evaluating how exaggerated traits evolve within and among populations (Bonduriansky 2007a; Shingleton and Frankino 2013; Pélabon et al. 2014; Voje 2016).

Sexual dimorphism is often viewed as the evolutionary divergence of males and females from a shared developmental architecture. However, dimorphism can arise through multiple pathways. Males and females may differ in average trait expression while retaining similar allometric relationships, or divergence may occur through changes in how trait size scales with body size. Recent evidence from the maritime earwig *Anisolabis maritima* demonstrates that females, like males, can exhibit positively allometric forceps, highlighting that weapon exaggeration is not necessarily restricted to males (Matsuzawa and Konuma 2025). Species expressing alternative male phenotypes provide an opportunity to examine these processes in greater detail because males themselves are partitioned into discrete reproductive forms. In such systems, one male morph may resemble females more closely than other males, raising the possibility that sexual dimorphism is concentrated within a subset of the male population rather than distributed uniformly across all males. Understanding whether alternative male phenotypes differ from females in scaling relationships therefore offers insight into how sexual dimorphism and trait exaggeration evolve from shared developmental programs. Under the developmental threshold

model, the major morph arises through activation of a developmental program for exaggerated trait growth when body size exceeds a critical value, whereas the minor morph retains a more “default” morphology produced in the absence of that program (Tomkins et al. 2005; Emlen et al. 2007); this framework provides a mechanistic basis for the classical view of males as exaggerated females (Darwin 1871; Bonduriansky 2007b) and predicts that the minor morph should resemble females more closely than the major morph in scaling relationships as well as in absolute trait size.

Species expressing alternative reproductive phenotypes provide particularly valuable models for examining the structure of allometry. In these species, males adopt discrete morphs that differ in weapon size, behaviour, and mating reproductive tactics (Gross 1996; Shuster and Wade 2003; Oliveira et al. 2008). Because morphs share much of their genome and developmental environment, differences in scaling parameters relationships can reveal how exaggeration varies evolves within a single species. However, empirical analyses of Although alternative male phenotypes have been documented in many taxa, relatively few studies have quantified morph-specific static allometry remain scarce allometries, and even fewer studies have investigated examined how environmental factors developmental environments influence both slopes and intercepts in polymorphic systems scaling relationships among morphs (Kochensparger et al. 2024; Kelly 2026).

Environmental conditions can influence trait and both body size variation and trait expression and consequently reshape may therefore alter static allometries (Emlen 1997; Frankino et al. 2005; Shingleton et al. 2007). Nutritional stress may, for example, can reduce expected trait size at a given body size, leading to lower intercepts investment in sexually selected structures, whereas

social ~~factors~~conditions such as rearing density ~~could alter patterns of trait expression relative to~~
~~body size~~may modify developmental trajectories through competition or social interactions.
 Environmental effects on scaling exponents would indicate plasticity in the relationship between
 trait size and body size, whereas ~~consistent slopes~~similar scaling exponents across environments
 would imply ~~canalisation~~canalization of ~~scaling~~allometric relationships (Frankino et al. 2005;
 Shingleton et al. 2007; Nijhout and German 2012). Determining whether ~~environmental variation~~
~~modifies intercepts, slopes, or both~~developmental environments modify scaling relationships is
 therefore ~~crucial~~important for ~~evaluating~~understanding the ~~flexibility of~~extent to which trait
 exaggeration ~~is environmentally sensitive.~~
 Complementary structural modelling of the same
 study system demonstrates that ~~environmental effects on~~developmental environments influence
 forceps ~~size are transmitted~~expression both indirectly ~~via~~through body size and directly through
 trait-specific ~~processes~~developmental pathways, particularly in males (Ozkan and Kelly
 submitted).

Beyond sex, morph, and environment, variation among maternal families ~~of related individuals~~
~~offers insights~~can provide insight into the developmental and evolutionary structure of allometry.
Families may differ not only in average trait expression but also in the strength of the relationship
between trait size and body size. Such variation could arise from genetic differences, maternal
effects, or shared developmental conditions. Quantifying among-family variation in allometric
slopes therefore offers a means of assessing whether scaling relationships. ~~Intercept and slope~~
~~may covary across lineages if shared genetic or developmental factors influence both expected~~
~~trait size and the strength of covariation with body size.~~ themselves vary among related
individuals (Lande 1980; Hansen and Houle 2008; Klingenberg 2008; Pélabon et al. 2014). ~~A~~

positive intercept-slope covariance indicates that families producing relatively larger weapons at average body size also exhibit steeper scaling relationships, reflecting coordinated variation in trait expression (Bonduriansky and Day 2003; Hansen and Houle 2008; Pélabon et al. 2014). Conversely, independence between parameters suggests modularity or relative decoupling of these components.

The European earwig (*Forficula auricularia*) provides a ~~powerful~~useful system for ~~integrating these perspectives on static allometry, examining how sexual dimorphism and alternative male phenotypes are structured through allometry.~~ Males express two discrete forceps morphs ~~that vary considerably in size and curvature,~~ commonly referred to as brachylabial and macrolabial males, which differ markedly in weapon size and brachylabial form~~sshape~~ (Simmons and Tomkins 1996; Tomkins et al. 2005) (Figure 1). Forceps function both as sexual weapons during male-male interactions and as defensive structures, making them traits with dual ~~roles~~functions (Kamimura 2014; Meunier 2024). Because these morphs coexist within the same population, earwigs provide an opportunity to ask whether males share a common scaling relationship that differs from females, or whether any divergence is associated more strongly with one male morph. Previous ~~work has~~studies have documented morph-specific differences in forceps ~~exaggeration and scaling,~~ raising the possibility that alternative male phenotypes may follow different allometric trajectories (Simmons and Tomkins 1996; Tomkins et al. 2005). Moreover, recent work on another earwig species revealed positive allometry in female forceps, suggesting that divergence occurs not only in absolute size but also potentially in scaling relationships. Nonetheless, most studies have primarily compared static slopes, sex differences in forceps scaling may be evolutionarily labile (Matsuzawa and Konuma 2025). In *F. auricularia*, the

macrolabial morph bears the more exaggerated weapon, whereas the brachylabial morph is more common in many populations and tends to be favoured under poorer developmental conditions. This pattern is broadly consistent with ~~less attention to intercept divergence, environmental sensitivity of scaling parameters, and variation among families; the possibility that brachylabial males retain a more female-like morphology under a threshold model of morph expression~~ (Tomkins 1999; Tomkins and Brown 2004; Tomkins et al. 2005).

Here, we quantify ~~sex-specific and morph-specific~~ static allometries of forceps length in female, brachylabial male, and macrolabial male *F. auricularia* reared under experimentally manipulated diet and density. ~~Rather than treating the scaling exponent as fixed, we test whether it varies across biological levels and environmental conditions. Specifically, we investigate whether sexual dimorphism in weapon exaggeration results from divergence in the scaling exponent, intercept, or both; whether macrolabial and brachylabial males differ mainly in expected trait size or in the strength of covariation between forceps size and body size; and whether larval environmental conditions influence intercepts, slopes, or their covariance. Finally, by estimating variation among families in intercepts and conditions. We first test whether the three phenotypic groups differ in allometric scaling relationships and whether any divergence is consistent across developmental environments. We then assess whether developmental conditions modify scaling exponents, we assess within each group and whether these components of scaling allometric relationships covary across shared backgrounds. Through vary among maternal families. By integrating sex, morph, environment, and family component effects within a unified hierarchical framework, we evaluate whether static allometry is best characterised as a stable descriptor or as an emergent property of multiple sources of variation (see also Simmons and Tomkins 1996;~~

~~Tomkins et al. 2005; Kelly 2026). weapon allometry is broadly conserved across phenotypic groups or whether divergence is concentrated within particular male morphs.~~

Methods

Study species and experimental design

~~Female (n = 115).~~ Adult female earwigs morphologically assignable to the *Forficula auricularia* species complex (*F. auricularia sensu lato*) were ~~hand-~~collected ~~by hand~~ at night from vegetation in Parc Outremont (Montréal, ~~QC~~Québec, Canada) during September and October 2024.

~~Individuals~~ The European earwig is now recognized as a complex of closely related lineages that cannot always be distinguished reliably using external morphology alone (González-Miguéns et al. 2020). Because individuals were not genetically assigned to lineage, we refer to the study population as *F. auricularia sensu lato* here and use *F. auricularia* for readability elsewhere in the manuscript. This taxonomic uncertainty should not affect treatment comparisons in the present study because all individuals originated from the same local population and were randomly assigned to experimental treatments, although caution may be warranted when comparing results among studies conducted on genetically characterized lineages.

~~Females (n = 115)~~ were transported to the laboratory at the Université du Québec à Montréal and ~~placed~~~~housed~~ individually in 90-mm Petri dishes ~~filled with~~~~containing~~ moist sand. Each dish was ~~provisioned~~~~supplied~~ with a cotton-plugged water tube, and TetraMin Tropical Flakes (United Pet Group, ~~Ine,~~ Germany), and ~~each female was~~~~females were~~ assigned a unique ID ~~code.~~~~identification codes.~~ Earwigs were maintained in ~~the dark~~~~darkness~~ at 10-°C ~~in an incubator~~ (Percival Scientific, Perry, IA, USA) to ~~mimie~~~~simulate~~ overwintering conditions. Dishes were

checked daily for egg laying, and food and water were replenished as ~~needed. Food and water~~
~~were removed at the time of egg laying~~required. Following oviposition (approximately early
 December), ~~after which food and water were removed and~~ dishes were ~~checked~~inspected weekly.
~~Starting~~Beginning in January 2025, dishes were checked daily for ~~egg~~-hatching. ~~The date of egg~~
~~hatching was recorded; hatching~~Hatching occurred between 13 ~~January~~ and 29 January 2025, and
 all 115 females successfully produced ~~hatchlings~~offspring.

For two weeks following hatching, families were ~~provisioned~~provided with ~~a food and water tube~~
~~and food (TetraMin Tropical Flakes: crude protein: 46%, crude fat: 11%, crude fibre: 3%). This~~
~~interval~~ad libitum. This period allowed nymphs to complete ~~an early~~ moult and ~~reach~~attain a
 size ~~sufficient to withstand~~suitable for manual transfer to experimental dishes. To manipulate ~~the~~
 developmental ~~environment~~conditions, we varied ~~two ecological factors: both~~ nutritional
~~condition (diet quality)~~ and social density. Diet treatment consisted of ~~two levels: either~~ a “good”-
~~quality~~ diet (100% TetraMin Tropical Flakes), ~~providing high nutritional quality, and a “) or a~~
~~poor”-quality~~ diet (75% TetraMin Tropical Flakes, ~~and~~ 25% non-digestible ~~α~~ -cellulose),
~~providing reduced nutritional content.~~ Density treatment consisted of ~~three levels: low density~~
~~(one individual, four, or eight individuals~~ per Petri dish), medium density (4 individuals), and
 high density (8 individuals). These treatments allowed us to examine both nutritional and social
 environmental effects on growth and morphology. ~~We used~~dish. Offspring from each family
~~were randomly assigned among treatment combinations using~~ a split-brood ~~experimental~~ design
~~in which offspring were randomly assigned to diet and density treatments within maternal~~
~~families to minimise~~to minimize confounding between maternal background and environmental
 treatment. Experimental dishes were maintained at 20°C ~~to mimic post-winter developmental~~

~~conditions~~ and contained a water tube, treatment-specific food, and a refuge (~~constructed from~~
 four ~~small pieces~~sections of paper straw ~~glued together~~). ~~A. In~~ total ~~of~~ 1,265 ~~Petri~~ dishes were
 prepared. ~~Females (mothers)~~established. Mothers were euthanized by freezing at -20°C after
 offspring had been assigned to treatments. Dishes were ~~checked~~inspected several times per week
 for mortality and ~~sexual maturation~~adult emergence, and food and water were replenished as
 needed.

Morphological measurements

Once offspring ~~eclosed to~~reached adulthood, they were removed from their ~~Petri~~ dish, ~~the date~~
~~was recorded, and they were~~ assigned a unique ~~ID~~identification code, ~~and the date of adult~~
~~emergence was recorded.~~ Pronotum length (a proxy for ~~overall~~ body size) and forceps length
 were photographed under a Leica S6D ~~stereo microscope~~stereomicroscope (Leica Microsystems
 Inc., Concord, ON, Canada) at $0.68\times$ magnification using Enersight software, ~~which~~
~~automatically adds a scale bar to each image.~~ Measurements were ~~taken~~obtained to the nearest
 0.001 mm (~~following Lieshout and Elgar 2009~~) using Fiji image ~~analysis~~ software (Schindelin et
 al., 2012). ~~All measurements~~ following established procedures (van Lieshout and Elgar 2009).
Measurements were ~~taken~~conducted blind to treatment ~~where~~whenever possible. To assess
measurement repeatability, a subset of 30 individuals was measured twice. Intraclass correlation
 coefficients indicated high repeatability (~~for both~~ pronotum ~~width-length~~ (ICC = 0.94;) and
 forceps length (ICC = 0.97). ~~Log-transformed pronotum width was mean-centred before analysis~~
~~to facilitate interpretation of intercepts and reduce collinearity among predictors.~~

Data preparation and statistical analyses

All analyses were conducted in the R Statistical Environment [R version 4.4.2; (R Core Team (2025))]. Data. Individuals were processed using functions from the tidyverse (e.g., dplyr), including recoding, filtering, and grouping operations. We retained individuals from experimental dishes that met predefined only when the final number of survivors preserved the intended density criteria to preserve the integrity of density treatment treatment (low density: one survivor; medium density: ≥ 3 at least three survivors; high density: ≥ 6 at least six survivors). Individuals lacking sex information were excluded. Forceps Final sample sizes for each phenotypic group and developmental environment are provided in Table S1. Pronotum length and pronotum width forceps length were log-transformed prior to analysis to linearize allometric relationships and reduce heteroscedasticity. Log-transformed pronotum length was not centred prior to analysis. As a result, model intercepts represent expected forceps length when log-transformed pronotum width equals zero and are therefore not directly interpretable as trait values at a biologically typical body size. Good diet and low density were treated as reference levels throughout. Female sex and brachylabial males served as reference categories where appropriate. Maternal identity was included as a random effect (grouping factor) in all models, with both random intercepts and random slopes for body size.

Male morph classification

Male morphs (brachylabial and macrolabial) were classified using finite mixture modelling implemented in the R package *mixsmsn* [version 1.1.12; (Prates et al. (2013))]. We modelled the distribution of raw. Raw male forceps length lengths were modelled as a two-component skew-normal mixture using the function *smsn.mix()*. Forceps length was analysed without prior size

correction so that morph classification reflects emergent weapon phenotype rather than residual variation after accounting for body size. Models were estimated using the expectation–maximization algorithm with automatic initialization (maximum 1000 iterations). Support for bimodality was evaluated using AIC and BIC. Posterior probabilities of component membership were calculated for each male, and individuals distributed, and males were assigned to morphs based on maximum the component with the highest posterior probability. All individuals were retained for subsequent analyses, avoiding arbitrary thresholds and preserving variation near the morph boundary.

Sex-specific allometry

We analysed sex-specific and morph-specific allometries using two complementary models to address distinct biological questions (see Supplementary Material for statistical model equations). The sex-specific model evaluates overall differences between females and males in scaling relationships and their environmental sensitivity. Because males comprise This procedure identified the two discrete morphs that differ in trait expression, this analysis effectively captures average differences between females and the male population as a whole, while allowing slopes to vary across predictors through interaction terms. The morph-specific model, restricted to males, explicitly tests for differences between brachylabial and macrolabial individuals in scaling parameters and their environmental sensitivity. This two-model approach therefore separates broad sex-level comparisons from within-male heterogeneity, avoiding the need to impose a single structure that simultaneously accommodates sex differences and morph variation within males. Results from the sex-specific model should thus be interpreted as averages across male

~~morphs, whereas the morph-specific model resolves variation within male morphs traditionally~~
~~recognized in *F. auricularia*: brachylabic and macrolabic males.~~

~~To quantify sex differences in allometric scaling and their environmental sensitivity, we fitted~~
~~We analysed allometric relationships using a hierarchical GaussianBayesian model using fitted~~
~~with the function brm() from the brms package (via brms (Bürkner 2017) and the cmdstanr~~
~~backend) [version 2.23.0; Bürkner (2017)]. Log-transformed forceps length was specified as the~~
~~response variable. Fixed effects. The final analysis included log 1,914 individuals from 110~~
~~maternal families and 667 rearing dishes. Log-transformed forceps length was modelled as a~~
~~function of log-transformed pronotum length (body size), sexphenotypic group, diet, density, and~~
~~all corresponding lower-order and higher-order interactions. among these predictors. Phenotypic~~
~~group comprised three levels: females, brachylabic males, and macrolabic males. Treating~~
~~females and the two male morphs as equivalent levels of a single factor allowed direct~~
~~comparisons among all phenotypes without assuming a single male allometric relationship.~~

~~Using descriptive variable names, the primary linear model can be represented in R notation as:~~

~~$\log F \sim \log P * \text{group} * \text{diet} * \text{density} + (1 + \log P \mid \text{mother id}) + (1 \mid$~~
 ~~$\text{rearing dish})$~~

~~where logF is log-transformed forceps length, logP is log-transformed pronotum length, group~~
~~is phenotypic group, mother_id identifies maternal family, and rearing_dish identifies the~~
~~rearing dish. Interactions betweeninvolving body size and other predictors allow theallowed~~
~~allometric slope (scaling exponent)slopes to vary across sexes and environmental conditions.~~
~~Main effects and lower-order interactions describeamong phenotypic groups and developmental~~

environments, whereas terms not involving body size described differences in model intercepts; however, because body size was not centred, intercepts correspond to trait values at log body size equal to zero and are not directly biologically meaningful. Three-way and four-way interactions involving body size were included to capture potential sex-specific environmental modulation of scaling relationships among groups and environments. Maternal identity ($n = 110$ families) family was included as a random varying intercept and random varying slope for body size, allowing families to vary in both intercepts and slopes differ in both average forceps expression and allometric scaling. Rearing-dish identity was included as a varying intercept to account for non-independence among individuals raised in the same dish.

Morph-specific allometry

To examine scaling architecture within males, we fitted a second hierarchical model restricted to males using brm(). Log-transformed forceps length was modelled as a function of body size, male morph (macrolabial versus brachylabial), diet, density, and all corresponding interactions. Body size $\times$ morph interactions quantify differences in scaling exponents between morphs, while lower-order terms describe differences in intercepts. Higher-order interactions with environmental variables allow both slopes and intercepts to vary across conditions in a morph-specific manner. As morph classification was based on absolute forceps expression, this analysis evaluates differences in scaling architecture among emergent weapon phenotypes without imposing size correction prior to morph assignment. Maternal identity ($n = 104$ families) was again included as a random intercept and slope for body size.

Priors and model fitting

Models were fitted in a Bayesian framework using brms. To evaluate whether allometric relationships deviated from linearity, we fitted an alternative model that included a quadratic body-size term and its interactions with phenotypic group, diet, and density. This model was specified as:

$$\log F \sim (\log P + I(\log P^2)) * \text{group} * \text{diet} * \text{density} + (1 + \log P \mid \text{mother id}) + (1 \mid \text{rearing dish})$$

Predictive performance of the linear and quadratic models was compared using approximate leave-one-out cross-validation (LOO) (Vehtari et al. 2017). Because the primary objective was to estimate interpretable condition-specific allometric exponents, the linear model was used to summarize average slopes, whereas the quadratic model was treated as a sensitivity analysis for localized departures from linearity.

Weakly informative priors were specified as $\text{normal}(0, 1)$ for fixed-effects and variance components to regularize estimates while remaining non-restrictive on the log scale. Models were fit using effect coefficients, $\text{normal}(0, 2)$ for the intercept, Student- $t(3, 0, 1)$ for residual and varying-effect standard deviations, and LKJ(2) for the maternal intercept-slope correlation. Models were fitted using four Hamiltonian Monte Carlo via cmdstanr chains of 4,000 iterations each, including 2,000 warm-up iterations, with an adapt_delta of 0.97. Convergence and sampling adequacy were assessed using standard diagnostics (Rtrace plots, effective sample sizes, and trace plots divergent transitions, and the potential scale reduction statistic [$\hat{R}$; Vehtari et al. (2021)]). Both models showed satisfactory convergence (Table S2).

Estimation of scaling exponents and plasticity

Condition-specific allometric slopes were ~~derived~~estimated from ~~the~~ posterior ~~draws~~distribution ~~of the linear model~~ using the function `emtrends()` ~~from~~in the package ~~emmeans~~ package ~~[version 2.0.1; (Lenth et al. (2019))]. Environmental plasticity in scaling was quantified as deviations in slope from the reference condition (good diet, low). Slopes were estimated separately for each combination of phenotypic group, diet treatment, and density)- treatment. Pairwise contrasts among phenotypic groups, diet treatments, and density treatments were derived from the same posterior distributions. Contrasts were calculated as the first level minus the second level—for example, female minus macrolabid male, good minus poor diet, or low minus medium density—such that negative values indicate a larger slope estimate for the second level. We report posterior medians and 95% credible intervals for all slopes, intercepts, and their contrasts. To aid interpretation, we also calculated posterior probabilities that slope differences exceeded zero and that their absolute magnitude exceeded 0.1 on the log-log scale. All inference is throughout. Inference was based on posterior distributions rather than null-hypothesis significance testing.~~

Results

~~Forceps length scaled positively with pronotum length across all individuals, confirming overall positive allometry (Table S1). Descriptively, females~~ Females exhibited a continuous distribution of weapon sizes across body sizes ($\bar{x} \pm \text{SE}$ pronotum length: mean $\pm$ SD = 1.77 ± 0.00514 mm; $\bar{x} \pm \text{SE}$ forceps length: mean $\pm$ SD = 3.59 ± 0.00824 mm, $n = 935$), ~~whereas males). Males~~ formed two discrete clusters corresponding to brachylabid ($\bar{x} \pm \text{SE}$ pronotum length: mean $\pm$ SD = $1.66 \pm$

0.00613 mm; $\bar{x} \pm SE$ -forceps length: $\text{mean} \pm SD = 4.06406 \pm 0.01430$ mm, $n = 491$) and
 macrolabic morphs ($\bar{x} \pm SE$ -pronotum length: $\text{mean} \pm SD = 1.80 \pm 0.00612$ mm; $\bar{x} \pm SE$ -forceps
 length: $\text{mean} \pm SD = 5.93 \pm 0.03372$ mm, $n = 488$; Figure 2).

Sex differences), with overlap in scaling relationships body size but substantial separation in
 forceps length (Figure 2).

**Allometric slopes differed strongly between sexes. Because males comprise two morphs with
 differing scaling relationships, sex-specific results should be interpreted as averages across
 male phenotypes. Females showed a moderate positive relationship between body size and
 forceps length ($\log P: \beta = 0.35$, 95% CrI: 0.07 – 0.65; Differences among females and male
 morphs**

Forceps length generally increased with pronotum length, and most posterior median slopes were
 below one (Figure 3a), whereas males exhibited substantially steeper scaling ($\log P \times \text{sex}: \beta =$
 0.61, 95% CrI: 0.25 – 0.98; Table S1). This divergence is evident in Figure 3, where male scaling
 relationships are consistently steeper than those of females across conditions. There was little
 evidence for consistent differences in intercept terms associated with sex ($\text{sex}: \beta = -0.06$, 95%
 CrI: -0.28 – 0.15). However, because body size was not centred, intercepts correspond to
 expected trait values at log body size equal to zero and are not directly biologically interpretable.
 Accordingly, differences between the sexes are best understood in terms of divergence in scaling
 slopes rather than shifts in intercepts.

Including a quadratic term for body size provided no meaningful improvement in model fit (see
 Supplementary Material for statistical model equation). Although the quadratic model showed a
 slightly higher expected log predictive density, the difference was small relative to its standard
 error ($\Delta \text{elpd} = 3.7 \pm 3.8$). In addition, credible intervals for the quadratic term overlapped zero,

indicating no clear evidence of curvature in the size–trait relationship. We therefore retained the linear model for all subsequent analyses.

; Table S5; Figure S1). Treating females, brachylabic males, and macrolabic males as three phenotypic groups revealed broadly similar scaling relationships across most developmental environments. Under good diet and low density, estimated slopes were 0.37 (95% CrI: 0.18–0.55) in females, 0.43 (95% CrI: 0.20–0.66) in brachylabic males, and 0.46 (95% CrI: 0.21–0.71) in macrolabic males. Pairwise differences among groups were generally small or uncertain, with contrast intervals overlapping zero across most environments (Table S6).

A quadratic model had somewhat better expected out-of-sample predictive performance than the linear model ($\Delta\text{ELPD} = 13.29$, $\text{SE} = 6.68$; Table S4), providing some evidence of curvature in the relationship between body size and forceps length. The quadratic predictions did not substantially alter the broad comparison among phenotypic groups, although they revealed localized departures from linearity (Figure S2). Curvature was most apparent near the extremes of the observed body-size range in several macrolabic male groups, where predictions were influenced by relatively few individuals. These edge patterns should therefore be interpreted cautiously and do not necessarily imply strong curvature across the central portion of the body-size distribution. We use the linear model to summarize average condition-specific allometric exponents and treat the quadratic model as a sensitivity analysis.

Across the full set of developmental environments, there was little evidence that females and brachylabic males differed consistently in allometric slope (Table S6). The contrast interval between females and brachylabic males included zero in every diet-by-density combination

examined. Macrolabial male slopes were also similar to those of females and brachylabial males under most environmental conditions. The clearest exception occurred under poor diet at medium density, where the estimated slope for macrolabial males (1.23, 95% CrI: 0.77–1.73) was steeper than those estimated for females (0.33, 95% CrI: 0.15–0.49) and brachylabial males (0.26, 95% CrI: 0.04–0.49). Pairwise contrast intervals excluded zero in this environment (Table S6). This result should nevertheless be interpreted cautiously because the estimate was based on only 18 poor-diet macrolabial males at medium density (Table S1).

Posterior median slopes for females and brachylabial males were below one in every developmental environment examined. The posterior median for macrolabial males exceeded one only under poor diet at medium density (1.23, 95% CrI: 0.77–1.73). Because this credible interval included one, the evidence that this relationship was hyperallometric rather than isometric was uncertain. The small sample of poor-diet macrolabial males further limits inference from this treatment combination (Table S1).

Environmental effects on scaling

~~Environmental effects on allometry~~ Estimated environmental differences in allometric slopes were generally weak, small or uncertain. In females, scaling relationships were largely parallel across diet and density treatments; contrast intervals included zero at all densities (Figure 3a), and interactions between body size and environmental variables had 95% credible intervals that overlapped zero (Table S1), indicating little evidence of environmentally induced changes in scaling exponents. In males, there was some indication that the nutritional environment may influence scaling relationships. The interaction between body size, sex, and diet was negative ($\log P \times \text{sex} \times \text{diet}$: $\beta = -0.68$, 95% CrI: -1.25 — -0.11), consistent with a reduction in the steep

male allometric slope under poor dietary conditions. This pattern is visible in; Table S7), providing little evidence that nutritional conditions altered scaling relationships. Diet contrasts for brachylabic males were likewise uncertain across all density treatments (Table S7).

The clearest indication of environmental variation involved macrolabic males, but these estimates should be interpreted cautiously because poor-diet macrolabic males were uncommon at low density ($n = 3$), medium density ($n = 18$), and high density ($n = 13$) (Table S1). At medium density, the estimated slope was steeper under poor diet than under good diet (good–poor contrast = -0.81 , 95% CrI: -1.33 to -0.29 ; Table S7). Diet contrast intervals included zero at low and high densities (Table S7).

Density contrast intervals also included zero for females and brachylabic males (Figure 3e, where slopes under a poor diet appear shallower than under a good diet, particularly at larger body sizes. However, credible intervals for several condition-specific slopes and derived contrasts overlapped zero (Table S1), indicating that support for diet-dependent changes in scaling is limited and should be interpreted with caution. By contrast, rearing density showed no consistent effect on scaling relationships. Although slopes varied slightly across density treatments (Figure 3), and interactions between body size and environmental variables had 95% credible intervals that overlapped zero (Table S1), indicating little evidence of environmentally induced changes in scaling exponents. Environmental effects on intercept terms were also limited and uncertain. Main effects of diet and density overlapped zero, and although there was weak evidence for a sex-by-diet interaction (sex $\times$ diet: $\beta = 0.29$, 95% CrI: -0.01 – 0.58), uncertainty in this estimate was high. As above, intercepts are not directly biologically interpretable because the body size predictor is uncentred.

Morph-specific patterns in males

Analyses separating male morphs (Table S2) showed clear differences in overall forceps size between morphs, with macrolabial males exhibiting larger forceps than brachylabial males (morph: $\beta = 0.26$, 95% CrI: 0.07–0.44). Allometric slopes were positive across morphs (logP: $\beta = 0.36$, 95% CrI: 0.14–0.59), but there was little evidence for consistent differences in scaling exponents between morphs (logP $\times$ morph: $\beta = 0.08$, 95% CrI: -0.24–0.40). Visual patterns suggest that macrolabial males may exhibit greater sensitivity of scaling relationships to diet (Figure 3c), whereas brachylabial males show more parallel scaling relationships across treatments (Figure 3b). However, statistical support for morph-specific differences in slope plasticity was weak, with credible intervals for interactions between body size, morph, and environmental variables overlapping zero (Table S2; Figure 4). Our results indicate that while male morphs differ clearly in overall trait expression, there is limited evidence that they differ consistently in the environmental sensitivity of allometric scaling.

; Table S8). Among poor-diet macrolabial males, the estimated slope at medium density exceeded those at low density (low–medium contrast = -1.26, 95% CrI: -2.26 to -0.29) and high density (medium–high contrast = 1.20, 95% CrI: 0.29 to 2.02; Table S8). Thus, the clearest estimated environmental differences were associated with poor-diet macrolabial males at medium density. Because all three poor-diet macrolabial groups had small samples, these contrasts should be regarded as provisional.

Among-family variation

~~Variation~~ Estimated variation among maternal families was ~~modest but non-zero~~ small (Table S4S3). The posterior median standard deviation of family-level intercepts was ~~low~~ ($\sigma_{\text{Intercept}} =$

0.03, 0.02 (95% CrI: 0.001—0.07107), indicating limited variation among families in overall trait values. There was somewhat greater among-family variation in forceps expression after accounting for body size, phenotypic group, diet, and density. The standard deviation of family-level allometric slopes among families ($\sigma_{\log p}$ = 0.05, (95% CrI: 0.004—0.126), consistent with 14), indicating some uncertainty and variation among families in the spread of slope estimates shown in Figure 4. relationship between body size and forceps length.

The correlation between family-level intercepts and slopes was negative but highly uncertain (r = -0.2179, 95% CrI: -0.91—0.7658), providing little evidence for a consistent association between intercepts forceps expression and allometric scaling among families. Variation among rearing dishes was small (SD = 0.01, 95% CrI: 0.00—0.02), as was residual variation (SD = 0.07, 95% CrI: 0.07—0.08).

Discussion

Static allometries are often treated as stable descriptors of morphological architecture (Huxley 1932; Gould 1966; Voje 2016), yet the extent to which scaling relationships vary among phenotypes and environments remains an important question in evolutionary biology (Pélabon et al. 2014; Voje 2016). By analysing females, brachylabial males, and macrolabial males as separate phenotypic groups, we found that forceps allometry in the European earwig was characterised more by similarity than by consistent divergence. There was no clear evidence that females and brachylabial males differed in slope in any developmental environment, and macrolabial male slopes were also similar to those of the other groups under most conditions. Where a clear slope difference was estimated, it involved macrolabial males under poor diet at medium density, a

treatment combination represented by only 18 individuals. Environmental contrasts were otherwise generally small or uncertain. These results suggest that forceps allometry is broadly similar across phenotypes and environments, while leaving open the possibility of limited, context-dependent divergence associated with the exaggerated male morph.

These findings differ quantitatively from two influential earlier analyses of *F. auricularia*. In two field populations from the Farne Islands, Tomkins and Simmons reported least-squares slopes of 0.68–0.70 for brachylabic males and 1.70–1.94 for macrolabic males, together with still steeper reduced-major-axis estimates, and concluded that the morphs differed in both slope and elevation (Tomkins and Simmons 1996). Tomkins et al. subsequently reported reduced-major-axis slopes of 1.15 for minor males, 3.95 for all major males, and 1.78 for asymptotic major males, interpreting the broken log–log relationship as evidence of size-dependent reprogramming (Tomkins et al. 2005). Our data likewise recover discrete male forceps modes, extensive overlap in body size, and a pronounced difference in average forceps expression, but they do not consistently recover steep macrolabic slopes: posterior medians were below 1 in nearly all environments, and morph contrasts were usually uncertain. Across studies, therefore, discontinuity in forceps expression appears more repeatable than strong divergence in within-morph allometric slope.

Direct numerical comparisons should nevertheless be made cautiously. The earlier studies analysed field-collected males from UK island populations, used pronotum width as the body-size measure, and classified morphs by forceps shape or forceps-length cutoffs; they also relied on least-squares or reduced major axis regressions. We analysed laboratory-reared offspring from a Montréal population of *F. auricularia sensu lato*, used pronotum length, classified males using

mixture modelling of forceps length, and estimated conditional slopes within developmental environments using a hierarchical Bayesian regression. In particular, reduced-major-axis slopes are not directly equivalent to the conditional regression slopes estimated here. Even with these differences, the qualitative contrast is substantial and suggests that steep macrolabic hyperallometry may vary among populations, lineages, or developmental contexts, or may be sensitive to how morphs and body size are defined.

Sexual dimorphism is classically framed as the divergence of males from a shared female “ground plan,” with males viewed as modified females and exaggerated male traits interpreted as departures from a default developmental programme (Darwin 1871; Bonduriansky 2007b). Under this view, alternative male morphs might differ from females in proportion to their degree of exaggeration. Our results provide only limited support for this expectation: brachylabic male slopes did not clearly differ from female slopes, whereas the clearest estimated departure involved macrolabic males in one sparsely sampled developmental environment. The principal distinction between the male morphs in our data was therefore one of forceps elevation and absolute expression rather than a consistently different scaling exponent. This distinction does not rule out a developmental threshold or reprogramming event, because such a transition could alter the elevation or timing of weapon growth without generating a uniformly steeper adult static allometry in every environment. However, because morphs were classified from the adult forceps distribution, the present analysis cannot independently demonstrate a developmental switch. Threshold models in horned beetles have likewise proposed reduced activation of weapon-growth pathways in minor males, although the extent to which alternative morphs are developmentally uncoupled remains debated (Tomkins et al. 2005; Emlen et al. 2007). Comparative evidence from

neotropical harvestmen suggests that minor-male weapon expression may track female expression more closely than major-male expression across species (Buzatto et al. 2014). Static allometries are often regarded as stable descriptors of morphological architecture (Huxley 1932; Gould 1966; Voje 2016). However, the evolutionary flexibility of scaling relationships remains debated, with some studies suggesting that slopes can evolve yet remain constrained by developmental integration (Harrison 2015). Our findings indicate that allometric slopes are largely stable but can vary modestly with environmental conditions, although this plasticity is uneven. In the European earwig, weapon exaggeration is not fully described by a single static slope, but instead shows patterns consistent with scaling relationships that differ between sexes and vary across developmental environments. Males exhibited steep positive allometry, consistent with expectations for sexually selected weapons (Kelly 2005; Kodric-Brown et al. 2006; Bonduriansky 2007a), and slopes tended to be shallower under nutritional stress. However, uncertainty in several condition-specific estimates indicates that support for slope plasticity is limited and context-dependent. Females, by contrast, showed comparatively stable scaling relationships across environments. Together, these results suggest that the allometric slope is largely stable but may exhibit modest environmental sensitivity, rather than representing a strictly fixed constant (Frankino et al. 2005; Shingleton et al. 2007; Nijhout and German 2012). Accordingly, our study provides evidence that condition dependence may extend beyond variation in mean trait expression to aspects of size-dependent exaggeration itself (Rowe and Houle 1996; Cotton et al. 2004; Tomkins et al. 2004; Pélabon et al. 2014).

The evolution of exaggerated traits is often interpreted through the lens of condition dependence and genic capture theory (Rowe and Houle 1996; Cotton et al. 2004; Tomkins et al. 2004), which

predicts heightened environmental sensitivity in sexually selected characters. Although many studies have focused on mean trait expression, condition dependence has also been interpreted as variation in allometric slopes (e.g. Bonduriansky 2007b; Vea et al. 2023). However, empirical tests of how environmental conditions shape scaling exponents remain relatively rare. If growth processes are condition-sensitive, the scaling exponent itself may vary across environmental contexts (Emlen 1997; Frankino et al. 2005; Shingleton et al. 2007; Nijhout and German 2012). Our findings provide limited evidence for such slope variation in a weapon trait. In males, the relationship between trait size and body size tended to be less steep under a poor diet, consistent with reduced exaggeration among larger individuals. However, uncertainty in several condition-specific estimates indicates that support for diet-dependent changes in scaling is uneven. In contrast, we found little consistent evidence that rearing density alters scaling relationships. Because the scaling exponent captures how disproportionately trait size covaries with body size, even modest or context-dependent variation in the slope could influence the strength of size-dependent exaggeration and therefore the opportunity for sexual selection to favour larger males. Accordingly, our results are most consistent with the idea that slope variation may be associated with environmentally mediated differences in growth processes, although the magnitude and generality of this effect remain uncertain.

Analyses separating male morphs refine the interpretation of morphological differentiation in earwig weapons. Macrolabial males exhibited the steepest allometric slopes under favourable conditions, but these slopes were reduced—and more variable—under a poor diet, suggesting that condition dependence may be more pronounced in this exaggerated morph. However, statistical support for morph-specific differences in slope plasticity was limited. In contrast, brachylabial

males showed more moderate slopes with comparatively little environmental sensitivity. Despite clear differences in trait size between morphs, we found little evidence that morphs differ consistently in intercepts independent of body size. Distinguishing between intercept and slope remains important because these parameters capture different statistical aspects of the relationship between trait size and body size: the slope describes how trait size covaries with body size, whereas the intercept reflects the expected trait size at a given body size (Bonduriansky and Day 2003; Pélabon et al. 2014). Our results therefore suggest that morph differences in exaggeration arise primarily from differences in scaling relationships rather than fixed shifts in expected trait size. This pattern is broadly consistent with models in which exaggerated phenotypes are more sensitive to condition, even when overall scaling relationships are similar (Klingenberg 2008; Hansen and Houle 2008; Buzatto et al. 2015).

Body size overlap between morphs also provides insight into the basis of morph determination. Although macrolabial males generally possess larger forceps, considerable overlap in body size occurs between morphs, with some small males expressing the exaggerated phenotype and many large males remaining brachylabial (see Figure 1). If morph expression depended solely on a strict body size threshold, a sharper transition would be expected. Instead, these observations suggest that body size alone is insufficient to predict morph identity. Previous work has shown that both environmental and genetic factors influence morph expression in *F. auricularia* (Tomkins 1999). Our findings are consistent with this view: morphs differ strongly in trait expression yet show broadly similar scaling relationships, indicating that morph differentiation does not require fundamentally different scaling patterns (see also Kelly 2024; Kelly 2026). Although we cannot estimate the heritability of morph identity directly, the overlap in body size between morphs and

the presence of modest among-family variation in scaling parameters suggest that morph determination is unlikely to be governed by a simple size-based switch.

This hierarchical structure—sex-specific differences in slopes combined with morph-specific differences in scaling sensitivity—suggests that exaggeration may evolve through modifications of scaling relationships across biological levels rather than through uniform shifts in trait expression. Sex-limited expression may allow selection to modify male size–trait relationships without directly affecting females, whereas morph differentiation appears to influence how scaling relationships vary across conditions. Selection may therefore act on different components of scaling across biological levels, particularly on slopes at the sex and morph level.

Among-family variation in scaling relationships was modest but non-zero, indicating that scaling parameters vary slightly across families with shared developmental or genetic backgrounds.

However, we found little evidence for a consistent relationship between family-level intercepts and slopes, as estimates of their covariance were highly uncertain. While family effects combine genetic and shared environmental influences, these results suggest that most variation in scaling relationships in this system is structured by sex, morph, and environment rather than strong divergence among families. Nonetheless, the presence of among-family variation indicates that scaling relationships are not fixed and may still provide a substrate for evolutionary change (Lande 1980; Pélabon et al. 2014).

In summary, these results support a view of static allometry as an evolvable and potentially environmentally responsive trait complex (Pélabon et al. 2014), although the strength of this responsiveness appears limited. Scaling exponents appear to be weakly condition-sensitive

descriptors of trait-body size covariation, particularly in males, but the magnitude and consistency of this effect remain uncertain. In contrast, there is little evidence for consistent effects on intercepts, suggesting that variation in exaggeration is more closely associated with differences in scaling than with shifts in expected trait size, albeit with substantial uncertainty in some estimates. Importantly, many condition-specific slope estimates were associated with wide credible intervals that overlapped zero, indicating that support for slope plasticity is context-dependent and should be interpreted cautiously. Accordingly, our results are most consistent with the view that scaling relationships are largely stable but may exhibit modest, environmentally contingent variation. The patterns observed here suggest that the evolution of exaggerated weapons may involve modification of scaling relationships across biological levels, rather than relying solely on changes in average trait size. Our analyses quantify statistical patterns of covariation and do not directly identify the developmental mechanisms underlying these relationships. Future work integrating quantitative genetic estimates of slope variation with experimental manipulations of selection will be essential for determining how readily scaling relationships evolve.

Whether a comparable developmental mechanism underlies the pattern in *F. auricularia* will require direct study of morph determination and forceps growth.

Morph-specific scaling does not follow a single pattern across male-dimorphic systems. In the Wellington tree wētā (*Hemideina crassidens*), weapon allometries did not align closely with the degree of weapon exaggeration or the presumed mating role of each morph (Kelly 2026). The present estimates provide a tentative contrast because the exaggerated macrolabial morph exhibited the clearest allometric departure, though only under one developmental condition.

Taken together, these studies suggest that alternative reproductive phenotypes can differ substantially in weapon expression without necessarily showing consistent differences in static allometry. The relationship between morph exaggeration and allometric slope may therefore depend on the developmental and functional context of each system.

The prevalence of posterior median slopes below one is also noteworthy. Sexually selected weapons are often expected to exhibit positive or hyperallometric scaling (Kodric-Brown et al. 2006; Bonduriansky 2007a), yet the posterior median slope was below one in all but one phenotype-by-environment combination. In the remaining combination, the median exceeded one, but its credible interval included isometry. This pattern is consistent with growing recognition that exaggerated sexual traits need not be hyperallometric (Bonduriansky and Day 2003; Eberhard et al. 2018; Kelly 2026) and with comparative work showing that allometric slope and intercept can evolve partly independently (Painting et al. 2024). Earwig forceps serve multiple functions, including male–male competition, predator defence, prey handling, and courtship interactions (Kamimura 2014; Meunier 2024). Such multifunctionality could limit disproportionate investment in a single function and favour more moderate scaling, although the present study does not test that mechanism directly. A recent study in *F. auricularia* also found no association between male forceps length and six behavioural or physiological traits commonly assumed to indicate male quality (Blackwell et al. 2024), suggesting that the selective information conveyed by forceps size may be less straightforward than a classical condition-dependent weapon model assumes. The broadly similar slopes estimated for females and both male morphs are consistent with shared functional or developmental constraints despite marked differences in absolute forceps size. Positive allometry in female *Anisolabis maritima*

(Matsuzawa and Konuma 2025) further illustrates that female forceps scaling varies among earwig species and should not be assumed to follow a single regime.

Environmental effects on scaling relationships were generally limited or imprecisely estimated.

Diet and density contrast intervals included zero for females and brachylabic males, and the clearest environmental contrasts were confined to poor-diet macrolabic males at medium density.

Because poor-diet macrolabic males were uncommon in all density treatments, these estimates are sensitive to small samples and should be treated as provisional. The quadratic model had somewhat better expected predictive performance than the linear model and revealed localized curvature in several relationships, especially near the extremes of the body-size distribution in macrolabic males. Those portions of the fitted curves were influenced by relatively few observations, however, and did not provide strong evidence of curvature across the central body-size range. We therefore interpret the linear slopes as average summaries of condition-specific scaling rather than complete descriptions of every relationship. Overall, the data provide little evidence for broad environmental restructuring of allometry, while suggesting a possible environment-specific response in the exaggerated morph that warrants targeted testing with more balanced samples. This interpretation is also consistent with evidence that selection on weapon allometry can act more strongly on the intercept than on the slope (Emberts et al. 2023), raising the possibility that developmental environments influence weapon expression primarily through average trait size rather than through large changes in scaling.

The similarity of female and brachylabic male slopes may provide some insight into morph differentiation, but it does not identify the underlying mechanism. Morph expression in *F. auricularia* is influenced by both genetic and environmental factors (Tomkins 1999), and morph

frequencies can shift with population density over short geographic distances (Tomkins and Brown 2004). The substantial overlap in body size between brachylabid and macrolabid males in our sample mirrors the earlier field studies and confirms that body size alone does not perfectly separate the morphs (Tomkins and Simmons 1996; Tomkins et al. 2005). At the same time, the combination of discrete forceps modes and generally similar within-morph slopes suggests that the main divergence in our population lies in the elevation of forceps expression rather than in wholesale restructuring of the relationship between body size and forceps length. This pattern is compatible with several developmental scenarios, including a threshold-like shift in weapon growth, but adult static allometries alone cannot distinguish reprogramming from continuous growth processes, changes in growth duration, or resource constraints (Tomkins et al. 2005; Emlen et al. 2007). Direct measurements across juvenile development would be needed to identify which mechanism produces the observed morph difference.

Estimated variation among maternal families was small, although the posterior distribution for the family-level slope standard deviation indicated some heterogeneity in scaling among families. The correlation between family-level intercepts and slopes was highly uncertain, providing little evidence that families producing relatively large forceps also tended to show steeper allometries. Because maternal identity combines genetic effects, maternal effects, and aspects of the shared early environment, these data cannot establish that the observed family variation is heritable. They instead indicate that scaling relationships may vary modestly among maternal lineages and motivate designs capable of separating genetic from environmental sources of that variation (Lande 1980; Pélabon et al. 2014).

More broadly, our results are compatible with viewing static allometry as a hierarchical phenotype shaped by variation at several biological levels (Emlen and Nijhout 2000). The most robust pattern in this study was broad similarity across females, male morphs, and developmental environments, accompanied by a limited number of context-specific departures. Analysing the male morphs separately was nevertheless important because pooling them would have concealed the steeper slope estimated for poor-diet macrolabid males at medium density and the greater curvature near the extremes of some macrolabid body-size distributions. Studies that pool alternative male phenotypes may therefore overlook heterogeneity in scaling, even when that heterogeneity is subtle or restricted to particular environments (e.g. Kelly 2005; see also Kochensparger et al. 2024). Our findings support treating alternative reproductive phenotypes explicitly in analyses of sexual dimorphism, while also emphasizing that differences among morphs should not be assumed to be pervasive. When considered alongside the earlier UK studies, the present results further suggest that steep major-morph hyperallometry is not necessarily an invariant species-level property but may depend on population, lineage, developmental environment, morph definition, or analytical scale. In *F. auricularia*, forceps allometry appears broadly similar across phenotypes and environments, with tentative evidence that the exaggerated male morph may deviate from this shared pattern under certain developmental conditions.

Data availability

Data are available through the OSF repository at https://osf.io/m7c53/overview?view_only=ce50e42b29284cbe8164e768ecb9c2b1.

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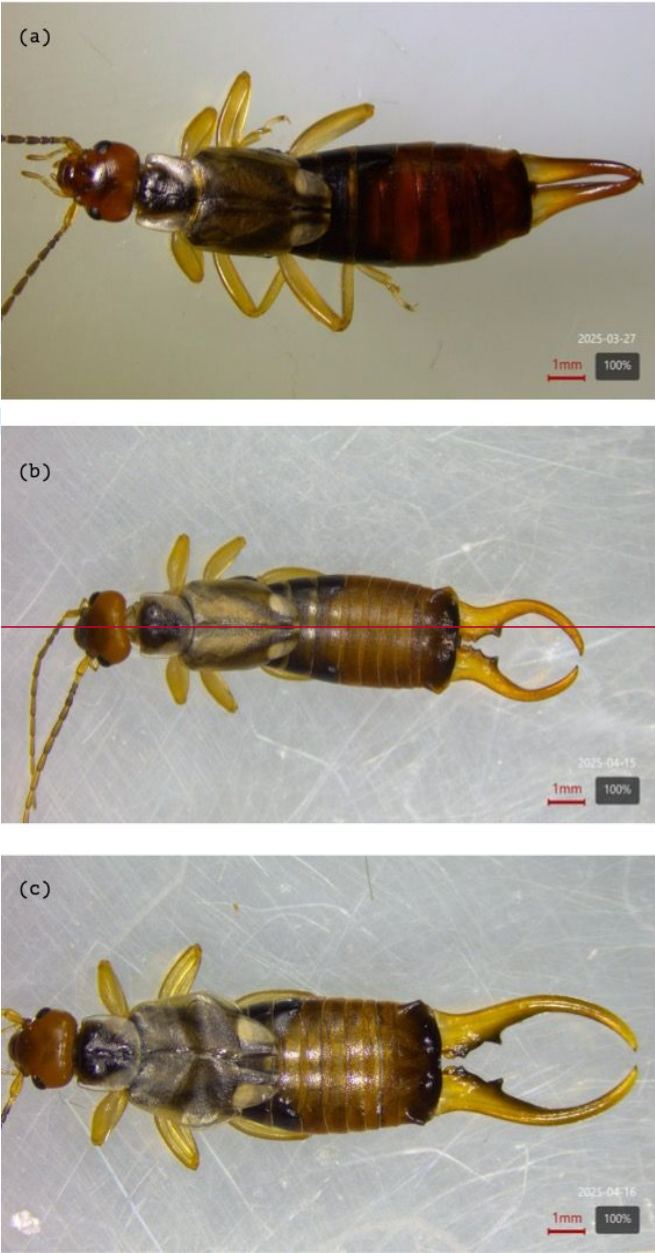
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894 **Figures**



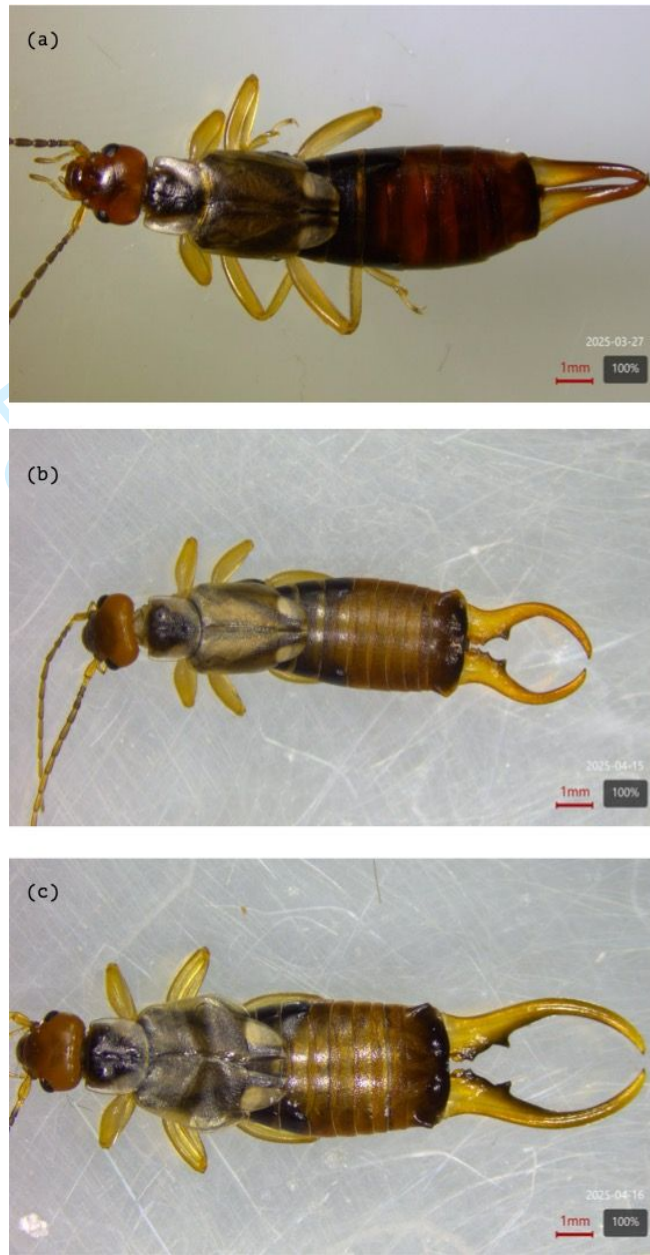


Figure 1: Visual comparison of forceps size and shape across sexes and male morphs: (a) female, (b) brachylabic male, and (c) macrolabic male. Macrolabic males display exaggerated have longer and more strongly curved forceps consistent with positive allometry, while than brachylabic males and females exhibit relatively reduced forceps. Photographs are of adults from the experimental population (Parc Outremont, Montréal). Scale bars = 1 mm.

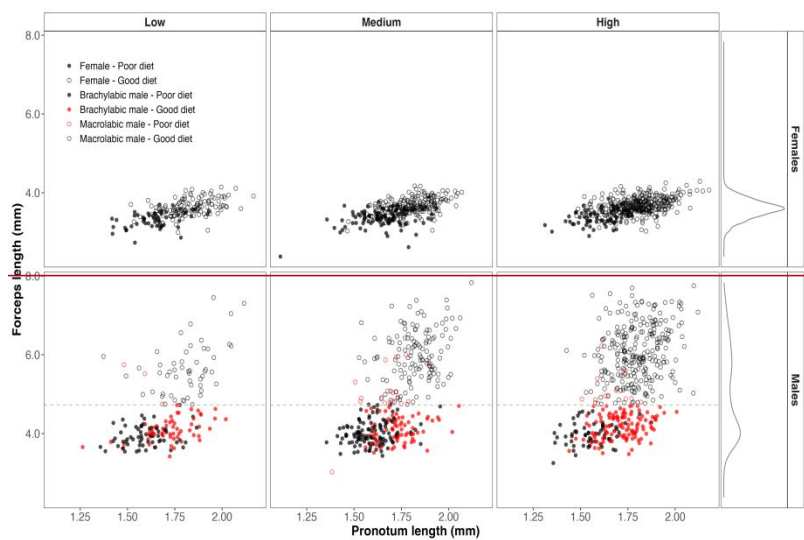


Figure 2: Static allometries of forceps length as a function of pronotum length across density treatments. Females (top row) and males (bottom row) are shown separately. In males, brachylabic (solid points) and macrolabic (open points) morphs are distinguished, with diet treatments indicated by shading. Males form two clusters corresponding to morphotype, with overlap in body size between morphs. The dashed horizontal line in the male panels indicates the forceps length threshold used for morph classification.

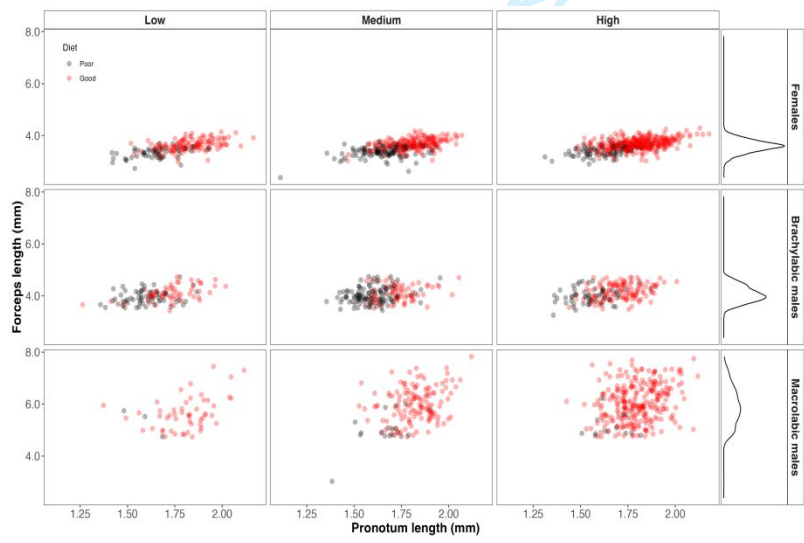


Figure 2: Relationship between pronotum length (body size) and forceps length in female, brachylabid male, and macrolabid male *Forficula auricularia* reared under two diet treatments (good and poor) and three density treatments (low, medium, and high). Points represent individual adults. Rows correspond to phenotypic groups and columns correspond to density treatments. Colours indicate developmental diet. Density distributions of forceps length for each phenotypic group are shown in the right-hand panels. Females exhibited a continuous distribution of forceps sizes, whereas males formed two discrete morphs that overlapped in body size but differed markedly in forceps length. Macrolabid males possessed substantially longer forceps than brachylabid males across the observed range of body sizes. Sample sizes for each diet-by-density combination are provided in Table S1.

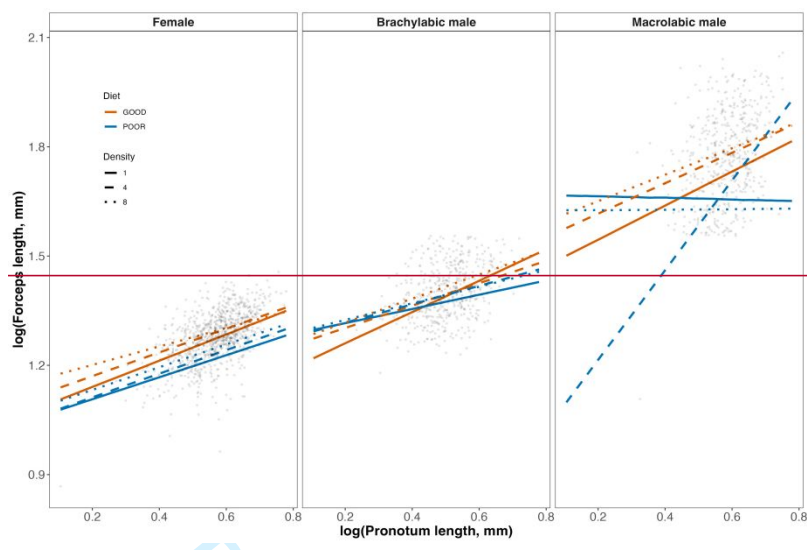


Figure 3: Predicted relationships between body size (log pronotum length) and forceps length across sexes and morphs for each environmental condition combination. Lines show posterior median predictions from the Bayesian model, and points represent raw data.

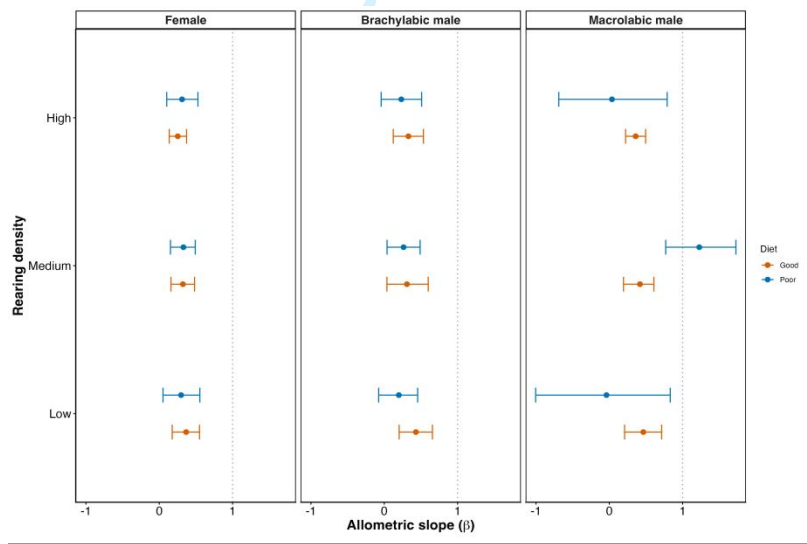


Figure 3: Posterior median allometric slopes (β) and 95% credible intervals describing the relationship between forceps length and pronotum length in females, brachylabic males, and macrolabic males across diet and density treatments. Slope estimates were derived from a single Bayesian mixed-effects model fitted to all individuals, with phenotypic group, diet, density, and their interactions with body size included as fixed effects. The vertical dashed line indicates isometry ($\beta = 1$). Posterior medians below or above one correspond to hypoallometric

or hyperallometric point estimates, respectively; credible intervals indicate uncertainty relative to isometry.

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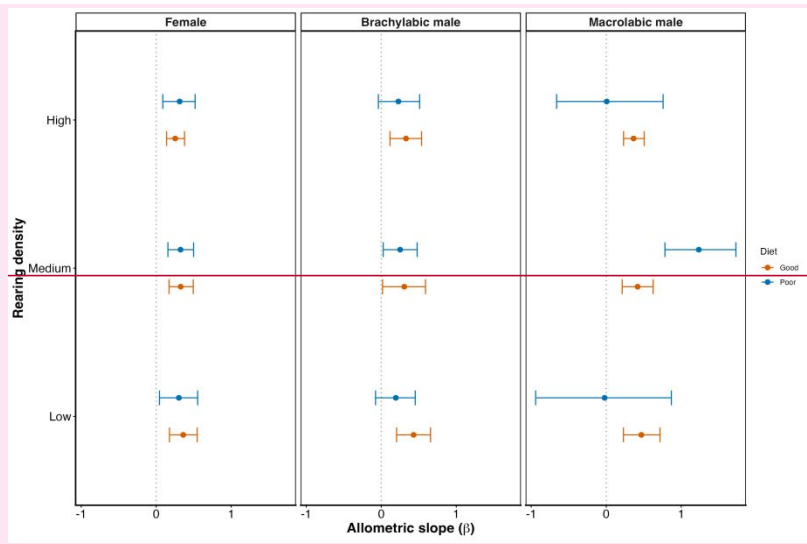


Figure 4: Posterior estimates of allometric slopes (β) across sexes and male morphs under different environmental conditions. Points show posterior medians and horizontal bars indicate 95% credible intervals. Estimates are shown for each combination of diet and density treatments.

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